GENERAL SPECIFICATION G4.47

Chemical Plant

CONTENTS

G4.47 CHEMICAL PLANT

G4.47.1 Definitions

(a) This specification shall cover the general minimum requirements of the chemical plants and systems which shall include the chemical storage tanks, solution preparation systems, water supply system, dilution, motive water, flushing and drainage systems, pumps and fittings, together with the system of administering the chemicals to the points of application.

(b) For any conflict between this specification and the particular specification or any other standard, the particular specification shall take the precedence and Contractor shall write to S.O. for clarification according to the procedure stated in the tender documents.

G4.47.2 Submittals

(a) The Contractor shall furnish complete fabrication, assembly, equipment and piping layout drawings, piping and instrumentation diagrams, and wiring diagrams, together with detailed specifications and data covering materials, parts, devices, and accessories for S.O.'s approval.

(b) The submittal shall be split up into different sections with a separate section for each chemical dosing system. Each system section shall be supplied with the following information:

(i) Chemical plant layout drawing;

(ii) Dosing equipment or skid fabricator's name, contact particulars, phone number, address, and email address;

(iii) Pump Manufacturer's Contact Name, Qualifications and Experience;

(iv) Technical schedule of all plant and equipment with associated fittings and valves;

(v) Chemical plant and skid piping and instrumentation diagrams with legend;

(vi) Equipment selection and datasheet which includes metering pump selections and performance curves, tank sizing, safety valves selection, etc.;

(vii) Metering pump appurtenances including calibration column and pulsation dampener (where required);

(viii) Control panel layouts and skid wiring diagrams;

(ix) Control narrative;

(x) Pressure relief valves and back pressure valves;

(xi) Spare parts; and

(xii) O&M manual.

G4.47.3 Chemical Plant and Ancillaries

(a) The Chemical Plant shall consist of but not be limited to the following items:

(i) Diaphragm metering, digital diaphragm of the necessary duty, standby and maintenance units;

(ii) Ball/diaphragm valves;

(iii) Check valves (where required in consideration of the type of dosing pump used);

(iv) Calibration cylinders;

(v) Pulsation dampeners (where required in consideration of the type of dosing pump type used);

(vi) Pressure relief valves;

(vii) Pressure sustaining valves;

(viii) Diaphragm seals;

(ix) Pressure gauges;

(x) Chemical Storage Tanks with all associated pipework and valving for fill, discharge, overflow, vent, drain, maintenance, instrumentation, interconnection and transfer;

(xi) Complete pipework, valving, pipe and valve supports, skid framework including pipe and valve elements as required for isolation, sampling and flushing applications; and

(xii) Local control panels and instrumentations.

G4.47.4 General

(a) The system shall be complete with chemical handling equipment, safety/personnel protection equipment (PPEs) and all other accessories, if specified and required to provide a comprehensive chemical plant in all respects.

(b) Each bund shall only serve one type of chemical and a common bund for more than one type of chemical is not allowed.

(c) The layout of the plant shall provide adequate space for access to carry out operations and maintenance of equipment.

(d) Installation arrangements for the pumps and associated pipework shall have unobstructed access to the controls, pump head, pump inlet and outlet check valves and sampling/flushing points.

(e) All valves shall be suitable for the intended design purpose and application. Where specific valve types are recommended by the Contractor, full documents shall be submitted for S.O.'s approval.

(f) All tanks and structure including all supports, the access platform and ladder walkway shall be designed and endorsed by the relevant discipline P.E.. The detailed designs shall be submitted to the S.O. for approval.

(g) The Contractor shall verify that each system component is compatible and consistent with all other chemical feed components on the project, that all pipe materials and sizes are appropriate, and that all devices necessary for a properly functioning system shall be provided.

(h) Devices and appurtenances necessary for a properly functioning system shall be constructed of materials consistent with the piping materials unless otherwise indicated.

(i) Similar components of different chemical feed systems shall be from the same Manufacturer to facilitate maintenance and stocking of repair parts. Whenever possible, identical units shall be provided.

(j) Unless otherwise specified, Health and Safety documentations, calculations and test procedures shall be furbished to support the requirements for every delivery of the chemical, and shall be subject to the approval of the S.O..

(k) All pipework supports and bolting accessories material of construction shall be stainless steel grade 316L, unless otherwise specified.

G4.47.5 Bulk Liquid Storage Tanks

(a) Storage tanks shall conform to the appropriate construction and welding standard, and shall be of enclosed vertical cylindrical construction, fabricated in either lined glass reinforced plastic or thermoplastic materials. The tanks shall be suitable for storing the solution at the concentration specified.

(b) Unless otherwise specified, vertical bulk storage vessels shall be provided as specified for the storage of chemicals received from bulk tankers.

(c) The storage tank shall be designed suitable for outdoor conditions.

(d) Unless otherwise specified, the minimum capacity of a single storage tank shall be the quantity of one delivery or one week's consumption at average demand, whichever is the greater. When more than one tank is specified, each tank shall accept at least the quantity supplied in one delivery or half a week's consumption at average demand, whichever is the greater. In addition, a free space of 10% of the tank volume shall be allowed in determining the gross capacity of each tank.

(e) For purposes of design, the maximum specific gravity shall be taken in fixing the mass and the minimum specific gravity in fixing the volume.

(f) Vertical liquid storage vessels shall be mounted above flood level on flat concrete plinths.

(g) All liquid storage vessels shall be located within dedicated bunds specific to each chemical and designed to hold 110% of the contents of the largest vessel:

  (i) Tank datasheets and specification shall be submitted with information of the tank construction, resin layer thickness (if applicable), fabrication materials, connections sizes, nozzle schedule, nozzle orientation, nozzle elevation, supporting saddles, required accessories, shop drawings and test certificates, etc. Drawings shall indicate connections and access manways locations and sizes. Drawings shall include a profile diagram of the entire tank wall lamination system indicating the thickness, resin designation, reinforcement and surfacing matt material of each layer including the structural and corrosion barrier layers.

(h) The connections shall include but not limited to the following:

  (i) Inlet;

  (ii) Outlet;

  (iii) Drain;

  (iv) Overflow;

(v) Cooling or heating coil (if necessary);

(vi) Vent; and

(vii) Contents indicator and/or transmitter.

(i) Usable capacity shall be measured from the invert of the tank overflow nozzles to the top of the pump suction nozzle.

(j) Tank shall be designed to withstand the hydrostatic head resulting from the tank being surcharged to 150 mm above the top of the tank.

(k) Flanges of tank nozzles shall be in accordance with BS 1092 PN10 minimum and nozzles shall extend at least 150 mm from outside face of tank to face of flange. Flanged nozzles shall be fabricated of the same material as the tank and shall be gusseted to the tank and provided with backing rings in stainless steel SS316L or otherwise reinforced in accordance with the governing standard.

(l) The filling connection shall incorporate a line isolating valve and a drain valve or quick coupling as specified. Filling pipework shall be 50 mm or 80 mm NB or any other size to suit in carbon steel to BS 3602 with 3 mm thick natural rubber or polyethylene lining or as specified. Filling pipes shall not be of plastic material of construction.

(m) Nozzles for drain connections shall be installed so the invert is flush with the bottom of the tank to allow complete draining of the tank.

(n) Tank level gauge shall be provided as magnetic level indicator type. Wet material shall be suitable for the process fluid and for the process conditions (temperature, pressure).. The exterior and housing materials shall be suitable for the environment. The gauge shall be provided with flushing and drain points for service with injurious or clogging substances.

(o) Unless otherwise specified, all tanks will be installed on concrete bases. The tanks will be anchored to the concrete base with suitable anchor bolts. The tank shall be provided with the appropriate number and size of lifting lugs for handling and installation and hold-down lugs for anchoring the tank to the concrete base.

(p) Bracketed flat surfaces shall be provided on the tank for installation of nameplate and certification plate.

(q) Glass reinforced plastic tanks (Includes fibre reinforced plastic tanks):

(i) Unless Otherwise specified, tanks shall be fabricated from uPVC or polypropylene lined glass reinforced plastic complying with AWWA D120 or BS EN 13923 and BS EN 13121 or alternatively from high density polyethylene. The exterior layer or body of the laminate shall be of chemically resistant construction. Design of glass reinforced plastic tanks shall include a cyclic loading equivalent to at least 7000 filling cycles over the design life.

(ii) Tank bottom shall have sufficient slope to prevent any possible sediment.

(iii) All tank nozzles shall be finished internally such that there are no exposed fibres.

(iv) Flange nozzle tolerances shall be in accordance to ASME RTP-1-2005. Flange nozzle offset from tank wall or top shall have a maximum permissible angular deviation of 1 degree from the nozzle face. Flange Flatness Tolerance shall be +/- 0.8 mm. Eccentricity between bolt holes and centre of nozzle shall be +/- 0.76 mm for 50 mm sized nozzles and +/- 1.52 mm for 80 mm sized nozzles or larger.

(v) All nozzle flanges shall have the minimum thicknesses and diameters that conform to the relevant governing standards.

(vi) Level sensor shall be hydrostatic type, bottle-side mounted. Refer to General Specifications G6 ICA for hydrostatic type level sensor.

(vii) The tanks shall be of hand lay-up, spray-up or filament-wound construction in accordance with the applicable governing standard. The finished laminate shall be constructed of a single generic type of thermoset resin and shall not contain colorants, dyes, fillers, or pigments unless otherwise specified.

(viii) Material of the resin shall be Bisphenol-A polyester or vinyl ester resins suitable for use with the specified chemical as recommended by the resin manufacturer. Material of the reinforcement shall be glass fibre with a suitable coupling agent. Surfacing mat shall be Burlington Formed Fabrics “Nexus Veil” or Nanofibers “Surmat 100”.

(ix) Unless otherwise stated, the inner surface layer of chemical storage tanks shall consist of a resin rich corrosion barrier with a minimum thickness of 110 mils. The surface of the corrosion barrier exposed directly to the corrosive chemical shall consist of a single-ply surfacing veil a minimum thickness of 10 mils. The remainder of the corrosion barrier shall consist of two layers or more of chopped strand mat or equivalent.

(x) The top of the tank shall be reinforced as specified in the applicable governing standard. Additional reinforcement shall be provided as necessary to support the required accessories and personnel loads. Tank roofs shall be provided with a nonslip finish over the entry roof surface as recommended by the tank manufacturer.

(xi) External coating of the tank shall contain UV inhibitor.

(r) Thermoplastic Tanks

(i) Tanks shall be fabricated from high density polyethylene (HDPE) or polypropylene (PP) and shall comply with BS EN 12573 Parts 1 & 2. Drilled flange connections and blank flange plates shall be fabricated to be resistant to corrosion from contact with small quantities of chemicals likely to be encountered during the design life.

(s) Venting and Overflow Arrangement

(i) Venting and overflow pipes shall be independent;

(ii) Vent shall be provided with a cowl or gooseneck and an insect screen;

(iii) A vent pipe of at least 150 mm in diameter shall be installed at the highest point in the tank roof and shall discharge to open atmosphere in a manner approved by the S.O.;

(iv) An overflow pipe of at least 100 mm in diameter shall be installed at a position of 350 mm–400 mm below the base of the vent pipe. The overflow shall discharge within the bund; and

(v) The vent and overflow pipes shall be properly supported and arranged so that they can be dismantled for inspection.

(t) Outlet Arrangement

(i) Each tank outlet connection shall be flanged, suitably gusseted and provided with a backing ring in stainless steel SS316L;

(ii) Two isolating valves shall be installed in series directly at the tank outlet; and

(iii) A separate drain branch with isolating valve shall be provided at the lowest point in the tank.

(u) Inspection Manhole and Cover

(i) Each storage tank shall have two 600 mm diameter bolted manways, one in the top and one in the side above bund flood level. The manways shall be provided with lifting davits to facilitate removal of covers;

(ii) Each manway shall be flanged, fully gasketed, and furnished with a fabricated blind flange having the same properties as the tank; and

(iii) Side access manways shall be cantilevered approximately 900 mm above the bottom of the tank.

(v) Ladders and Handrails

(i) An access platform shall be provided at the top of each storage tank complete with kick-plates and hand railing;

(ii) The platforms shall be supported independently of the tanks. Where more than one tank are installed, a walkway between adjacent vessels shall be provided also complete with kick-plates and hand railing;

(iii) Access to the top of the tanks shall be provided by means of an exterior ladder with safety cage and intermediate platforms where necessary. Tanks with a total height greater than 2000 mm from the finished floor shall be provided with a safety cage. The ladder shall be supported on and anchored to the concrete base and bracketed to the tank shell as needed; and

(iv) The ladder, access steelwork and hand railing shall comply with the General Specification on Metalwork and Workmanship requirements.

(w) Level Sensor

(i) Each tank shall be equipped with level transmitter. Level sensor shall be hydrostatic type or as otherwise specified in the particular specification which shall be compatible with the chemical. The output signal shall be transmitted to an indicator mounted adjacent to the filling point and as per specified location;

(ii) Each storage tank shall be equipped with a level transmitter. The output signal shall be transmitted to the SCADA system, to an indicator mounted locally to the tank and as per specified location;

(iii) In addition, level switches shall be provided to initiate high-high level alarms on the SCADA system and at the tank filling point. These switches shall be independent of the contents gauge;

(iv) Each tank shall be equipped with a high-level switch for initiating a high-level alarm at the tank-filling point and low-level switches for process control. The level switch shall be independent of the contents gauge / Indicator; and

(v) Level indicator/transmitter shall be provided for continuous level measurement and high level and low-level switches for the chemical tanks for alarms and feed pump control/trip for all required chemicals.

(x) Nameplates

(i) Each tank shall be provided with a nameplate. The nameplates shall be of stainless steel or white phenolic material with black engraved lettering 75 mm high and shall be mounted on the tank straight shell;

(ii) The chemical name and the tank tag number shall be engraved on the nameplate;

(iii) A stainless-steel certification plate shall be mounted below each storage tank's nameplate, containing information of the tank fabricator's name, date of manufacture, manufacturer's serial number, and resin designation for entire tank (structural and corrosion barrier), maximum allowable concentration and temperature of the specified chemical solution that can be stored safely; and

(iv) Material Safety Data sheet with cover shall be mounted onto tank.

(y) Protective Coating

(i) All nuts and bolts shall be manufactured from stainless steel 316; and

(ii) A selection of colour pigments available for the vessel shall be provided.

G4.47.6 Liquid Chemical Unloading/Filling Station

(a) Facilities shall be provided to receive supplies of chemicals by road transport and for unloading, transporting to storage areas or filling into storage tanks.

(b) Where chemicals are delivered in Intermediate Bulk Containers (IBC) or by road tankers, a bund shall be provided to accommodate the full length and width of the delivery vehicle and be aligned with all storage tanks' filling points. The bund may be formed from the kerb of the pavement and raised humps in the road surfaces.

(c) The bund shall be capable of holding liquid spillage of at least 1000 litres or the contents of the largest road tanker where applicable.

(d) The bund shall have drainage channels or gulleys which discharge to a sump with open mesh cover. The sump shall have an outlet valve discharging to foul sewer. The valve shall have an extension spindle and headstock equipped with a key-operated mechanical interlock. When the valve is open, the key shall be held captive. Only when the valve is fully closed whereupon the key shall be released to operate the tank filling valve. The sump outlet valve shall normally be padlocked in the open position to prevent unauthorised tampering or removal of the interlock key.

(e) The filling points for each chemical shall be clearly identified by a nameplate fixed firmly to an upstand local to the filling point. The plate shall be in stove enamelled aluminium and the letters shall be 150 mm high. If specified, a tank contents indicator with 300 mm dial or digital display of equivalent visibility showing both % full and tonnage shall be mounted at each filling point. If specified, an alarm panel with 75 mm diameter domed indicator lamps, audible alarm and acknowledgement/reset button all in a weatherproof enclosure shall be provided at each filling point.

(f) A 25-litre capacity plastic carboy shall be provided for each filling point to allow chemical residue to be drained from the filling pipe after a delivery. The carboys are to be stored inside the fill point kiosk.

(g) Each chemical storage tank shall be provided with a dedicated filling pipe which is not manifolded. The filling connection shall incorporate a line isolating valve, check valve and a drain valve.

(h) Filling pipework arrangement shall be designed to receive deliveries direct from road tankers by transfer pump or air padding at a pressure of 2.1 bar.

(i) To prevent unauthorised use, each filling pipe shall be provided with a lockable isolating valve or motorised valve or as indicated in the drawings.

(j) Each filling pipe inlet shall be a standard male quick coupler connection of nominal diameter size 50 mm or 80 mm matching the standard female snap-on coupler on the hose of chemical supply tankers. The filling pipe inlet male coupler shall be with matching snap-on dust cap attached with a chain to the coupler. For the filling pipes of sodium hypochlorite, ferric chloride and sodium bisulphite, the male coupler, dust cap and chain shall be of polypropylene (PP) material of construction. For all other chemicals not including sulphuric acid, the material of construction for these fittings shall be SS316L.

(k) For sulphuric acid, the filling pipe inlet shall be a flange and a lock shall be fitted through one hole in the flange. The flange shall conform to BS 10 Table E.

(l) The piping material of construction for sulphuric acid shall be selected taking particular consideration of the concentration and operating temperature of the chemical.

(m) Filling pipes shall be positioned in a vertical plane with the inlets facing the roadway and approximately 750 mm to 900 mm above ground level or other requirements to suit the local suppliers of the chemical. Filling pipe inlets shall be no more than 2.5 m from the delivery tanker outlet connection.

(n) The filling pipes for each chemical shall be brought to a single charging point and the charging points of different chemicals shall be segregated.

(o) The line downstream of the inlet connection, isolation valve and check valve shall rise vertically to a height slightly above the inlet to the tank, then slope downwards to the tank to ensure that it drains towards the tanks and it shall be securely supported to withstand vibration. The vertical section of each filling line shall have a 25 mm nominal diameter size drain leg with flexible tail tube within the kiosk.

(p) Each filling point shall be provided with a safety shower complete with an eye bath and a water supply with a hose for washing down any spillages and leakages during delivery. The hose shall be of reinforced plastic suitable for high pressure duties. The hose shall terminate in a nozzle and shall be complete with a hose reel and a suitable enclosure.

(q) All metal components of the filling system shall be equipotential bonded to the plant main earth terminal.

(r) Plastic filling pipes shall not be used. Where metallic piping material is unsuitable for any particular chemical, the appropriate interior lining material shall be used to lined metallic pipes taking into consideration the concentration and operating temperature range of the chemical.

G4.47.7 Liquid Chemical Unloading/Transfer Pumps

(a) Horizontal end-suction centrifugal pumps used for chemical transfer shall be of the back pull-out type, unless otherwise specified.

(b) Pumps shall be constructed in suitable chemical resistant thermoplastic or thermoset materials such as glass-filled epoxy resin or high-density polyethylene with external metal armouring.

(c) Road tanker discharge pumps where installed shall be rated at 20 m^3/h. Suitable priming and venting provisions shall be provided.

(d) The pumps shall be of the mechanical seal type. Alternatively, sealless magnetic-drive pumps shall be used.

G4.47.8 Solid Chemical Unloading and Storage

(a) Chemical Bulk Bags Unloading and Storage

(i) Facilities shall be provided for unloading chemicals delivered in individual bulk bags.

(ii) Chemical bulk bags shall be arranged on pallets in storage areas to provide access corridors for personnel and mechanical handling equipment.

(iii) The following features shall be considered in the layout of the storage areas:

1) Good access for mechanical handling equipment such as fork lift trucks, pallet trucks, trolleys, overhead monorail hoists, etc. Adequate access corridors shall be provided and shall be sized for manoeuvring fork lift trucks and other trucks into position making allowances for their turning radius;

2) Bulk bags shall be stacked on pallets shall be placed about 25 to 50 mm away from the walls, particularly if condensation on walls is likely to occur;

3) Good access to bulk bags and packages shall be provided to ensure good stock rotation; and

4) Adequate space shall be provided in storage areas to accommodate any listing of palletised crates.

(b) Bulk Bag Discharge Units

(i) Refer to the General Specification Bulk Bag Dischargers.

(c) Chemical Sacks Unloading and Storage

(i) Means shall be provided for unloading chemical supplied in sacks from suppliers' delivery vehicles and transferring the same to store;

(ii) Manual handling of packages within the store using sack trolleys or similar will be acceptable only when the maximum consumption does not exceed 100 kg/day or as otherwise specified. Any lifting of packages above floor levels shall be fully mechanised;

(iii) The material shall be stored in their original packages in a cool and dry area on pallets; and

(iv) All necessary steps shall be taken to alleviate hazards from chemical dust and to contain spillages during handling operations.

(d) Solid Chemical Handling

(i) Pallets

1) Pallets for transport and storage of chemicals and other specified material shall be of the close and open-boarded and reversible and non-reversible types with two-way and four-way entry, as specified;

2) In open boarded pallets, a maximum spacing of 50 mm between deck boards (whether top or bottom) shall not be exceeded;

3) Pallets dimensions shall be as specified and shall be fully compatible with the fork lift and pallet trucks supplied under the Contract;

4) Pallets shall be constructed in accordance with BS ISO 6780;

5) Deck boards shall have sectional dimensions not less than 25 x 125 mm and shall be of European redwood or approved equal. Blocks shall be 100 mm cubes and bearers shall have sectional dimensions 50 x 100 mm; and

6) Blocks and bearers shall be of European Oak or approval equal. Wood shall be fully treated against fungal, insect and animal attack. The Board name shall be burnt into pallets.

(ii) Fork Lift Trucks

1) Fork lift trucks shall be of the 'sit-down' driver type with two stage mast and standard free lift. Trucks employed solely for use externally shall be diesel driven, and trucks employed for internal use to any degree shall be of the battery-operated type. Full recharge of the batteries of the fork lift trucks shall be possible in less than eight hours;

2) The trucks shall have hydraulic lift, back and forward tilt, and manually adjustable lateral fork spacing, and shall be complete with overhead guard, load backrest, buffer and automatic battery charger;

3) Each truck shall be provided with a battery charge indicator and POWER ON light and also with a 'dead man' type brake;

4) The lifting forks shall be made as single piece forgings in steel to PD 970, Grade 150 M19 or Grade 150 M28. Each fork shall be clearly forged and finished with a smooth surface and adequately radiused corners;

5) Each truck shall be permanently and legibly marked with; and

- Identification number;
- Safe working load; and
- Where applicable, the tyre pressure.

6) Fork lift truck design and testing shall conform to BS ISO 22915, BS 4338, BS 5639 and BS ISO 22915.

(iii) Pallet Trucks

1) Pallet trucks shall be of the hand powered hydraulic type with wheels made of polyurethane;

2) When loaded to its maximum safe working load the lowest position of the pallet from the floor shall be not less than 100 mm; and

3) The lowering speed shall be infinitely variable by hand lever which shall be in the neutral position whilst manoeuvring.

(iv) Vacuum Bag Loaders

1) Vacuum bag lifters shall consist of an extendible flexible lift tube with manually directed suction operating foot. Operating vacuum shall be provided by an electrically operated vacuum pump. Raising and lowering of the suction foot shall be achieved by adjustment of the air flow in the lift tube; air flow shall be controlled by a trigger mechanism or similar hand operated device on the suction foot;

2) The unit shall be capable of lifting bags or drums of up to 50 kg. The suction foot shall be suitable for lifting paper, plastic, woven polypropylene or jute sacks as applicable. Precautions shall be taken to minimise wear at the attachment point of the suction foot to the lifting tube;

3) The lift tube shall be in reinforced plastic material and shall be capable of lifting bags to any height between 150 mm and at least 1400 mm above floor level unless otherwise specified. Lateral movement either side of vertical of at least 500 mm shall be possible when the tube is fully extended. PVC hosing with steel reinforcing wire shall be used for the suction tube connecting the vacuum pump to the lift tube;

4) When specified for installation on a support jib the design shall allow loads to be lifted and manoeuvred upwards, downwards, sideways or rotated through a minimum of 2700 for exact position requirement. Vacuum hose shall be provided with catenary supports. Jibs and associated stanchions, where specified, shall be of painted mild steel;

5) A safety release system shall be incorporated to ensure that loads are handled with complete safety in the event of a sudden loss of vacuum;

6) The vacuum generation system shall be powered by an electric motor protected to at least IP54. Precautions shall be taken to prevent excessive vacuum from harming the vacuum pump; and

7) The system shall be provided with an in-line cartridge filter system. A "bin" filter system, fitted with a primary cloth element and a secondary cartridge filter, shall be provided to protect the pump when the unit is to be installed in a particularly dust prone area.

(v) Roller Table

1) Chemical sacks shall be lifted by the vacuum bag loader and placed onto a roller table at approximately 1000 mm above floor level to facilitate manoeuvring of the sacks;

2) The roller table shall have rollers with roller diameters of at least 65 mm and the clearance between between rollers shall be at least 40 mm; and

3) Rollers shall have sealed-for-life bearings attached to a frame anchored onto the floor.

(vi) Inclined Belt Conveyor

1) An inclined belt conveyor shall receive the chemical sack from the roller table and convey the sack to the bag splitter;

2) The belt conveyor shall be complete with flat rubber belt, carrying idlers, return rollers, head and tail drums, side guards, loading end guards and adjustable support legs, etc., suitable for handling chemical bags of dimensions as specified;

3) The usable width of the belt conveyor shall be not less than 500 mm. The belting material shall comply with the requirements of BS EN ISO 14890 and the belt edge clearance shall be in accordance with BS 8438;

4) The conveyor shall be designed to contain all the chemical powder escaped from the chemical bags so that no chemical powder would fall from the conveyor belt onto the floor during operation;

5) The belt speed shall be designed to match the capacity of the bag splitting machine as specified. However, in no circumstances a belt speed greater than 18 m/min shall be accepted. In the design of belt speed, the inclination of the belt conveyor shall be considered, and slatted belt instead of flat belt shall be used if the inclination of the belt conveyor together with the designed belt speed may render the chemical bags to slide down or tip over from the conveyor;

6) Provision shall be made in the design of the conveyor for maintaining the belt tension automatically and the adjustment of the tail pulley of the conveyor to ensure the squareness of the belt. Intermediate guide rollers shall be provided to keep the belt in alignment;

7) Bearings of all rolling elements of the conveyor shall be of the sealed-for-life type; and

8) Each conveyor shall be driven by a geared electric motor which shall be integral with the head drum. The motor shall have a degree of protection of IP 55 to BS EN 60034-5 and its design shall be suitable for intermittent operation. A mechanical overload protection and anti-run-back device shall be provided.

(vii) Bag Splitter

1) Chemical sacks are fed to a bag splitter machine to have the sack slit and discharge the chemical powder into a hopper. The discharge chemical powder is then used to prepare the solution of the required concentration. The bag splitter shall conform to the following requirements:

- The splitting machine and accessories shall be of proven design with records of installation showing that they are free of operation and maintenance problems. The machine shall not be choked by small pieces of bag paper cut out during the splitting process.
- In general, chemical bags shall be loaded onto the roller table and shifted one by one to the inclined belt conveyor for automatic transfer to the splitting machine. The splitting machine shall then cut the chemical bags, sieve the bag fabrics, separate and discharge the chemical to a screw conveyor assembly. The screw conveyor assembly comprising horizontal (or slightly inclined), vertical and/or inclined screw conveyors shall transfer the chemical from the splitting machine into the chemical hoppers. Waste bags discharged from the splitting machine shall be compacted into a plastic sack for collection and subsequent disposal. The splitting machine and associated equipment shall be fully enclosed for dust containment during operation. A dust extraction unit shall be provided to prevent dust from flying out of the system during operation.
- The splitting machine and its associated equipment shall be fabricated from stainless steel grade 316L (termed as “stainless steel” hereafter).
- Conveyor loading tables compatible for mounting onto the bag splitter shall be used to transfer the powder bags at floor level to the bag splitter.
- Any lifting equipment such as hoists necessary to raise chemical bags from storage area floor level to tank charging level of the loading table shall be provided.
- The bag splitter shall be with a bag inlet, integral grinder, self-operating outlet of the flap type, level switch, replaceable dust extractor, dust filter with automatic pulse jet filter purge and waste bag compactor.
- Automatic bag splitters shall accept all bag sizes up to 1200 x 600 x 280 mm containing up to 50 kg material.

- Chemical sacks shall be introduced via an automatic infeed conveyor at a rate of up to 600 sacks per hour.
- Sacks shall be split by rotary blades which shall adjust automatically to the size of bags introduced to the splitter. Blades shall be selected for long life, taking into account the bag material type and the properties of the chemical being handled.
- Splitters shall be suitable for handling all types of paper, plastic and jute sacking materials.
- Splitters shall be designed to ensure that a minimum of 99% of bag contents is recovered from each bag. Discharged chemical shall fall through an outlet screen where extraneous material shall be removed.
- Empty sacks shall be discharged automatically into an integral compactor unit which shall extrude emptied and compacted bags into receiving bags.
- Geared drive assemblies shall be protected from contact with chemical being discharged so far as possible. Viewing windows shall be provided to allow inspection of the internals of the bag splitting section during operation.
- Bag splitters shall be designed to minimise dust escape into the atmosphere and shall be provided with integral dust control and pulse jet dust extraction systems. These shall be self-cleaning and shall return collected material to the discharge outlet from the splitter.
- Control units for each bag splitter shall be located adjacent to the loading station, readily accessible to the operator.
- All areas where dust is likely to be generated shall be provided with ambient air mechanical dust extraction/ filtration systems.
- From the bag splitter, the powder released from the sack shall discharge through an assembly configuration of flexible hose, pipe, nozzle and hopper. The unit shall be of stainless steel construction. A 'no-powder' alarm shall be provided which shall cut out the motor after three attempts.
- The feeding unit shall incorporate a screw feeder for accurately metering the powder to the wetting device. The screw feeder shall operate under control of an adjustable timer to provide a means of varying the batch concentration. The reproducibility of the feeder shall be better than + or - 3% by weight.
- All necessary means for unloading delivery vehicles, transferring the chemical into store, movement of the chemical into the charging area to charge into the bag splitter shall be provided.

(viii) Bag Splitter Dust Extraction Unit

1) Each bag splitting machine shall be equipped with its own dust extraction unit. The dust extraction unit shall be provided in such a way that no dust of appreciable concentration or quantity is discharged to the atmosphere. The filtered chemical dust shall be returned directly to the screening section for a built-in unit. Should an external unit be provided, a dust bin of not less than 50 litres shall be provided at the bottom of the dust extractor for the collection of filtered chemical dust;

2) The dust extractor shall be of stainless steel construction. The unit shall include an electrically driven dust extractor, chemical dust filter elements, side access doors and cable termination box. Dust extraction motor shall not be located inside the dust extractor;

3) The dust extractor shall be of side filter bag removal type. Removal of the filter assembly shall be effected from the front of the unit through a hinged access door. The filter assembly shall consist of flexible inserts which shall maintain the form of the element under dynamic air flow conditions;

4) Filter elements shall be made of suitable chemical resistant material and certified to BS EN 60335-2-69 Class L or better. Dust shall be removed from the filters by operation of a cleaning mechanism in the form of repulse air jets or vibration;

5) All electrical components shall have IP65 or better enclosures to BS EN 60529;

6) Solenoid valves shall be manifolded. Each unit shall be furnished with solid state sequencing timer and air pressure gauge. A differential pressure indicator and switch shall be provided to the compressed air pipelines feeding the extractor. A high differential pressure alarm shall be provided when cleaning or replacement of filter bag is required; and

7) If the connection between the dust extractor and the chemical handling equipment is by air duct, sizing of the air duct shall be designed to have the dust particles to travel at an optimum speed of not less than 18 m/s.

(ix) Bag Compactor

1) Each bag splitting machine shall be equipped with its own bag compactor to reduce the volume of the waste bags to less than one sixth of the original volume. The empty bags shall be transferred to the bag compactor automatically without manual operation;

2) The bag compactor shall be of mechanical type or hydraulic type. A discharge nozzle shall be provided for transferring and compacting waste bags into a plastic sack. The discharge nozzle shall be not less than 400 mm in diameter, fabricated from not less than 6 mm thick stainless-steel plate with a compactor transition section made of not less than 9 mm thick stainless steel plate; and

3) The bag compactor shall be mounted on the floor to facilitate operation and maintenance. The compactor housing shall be fabricated of structural steel and steel panels. A bin with castors shall be provided at the bottom of the bag compactor to collect the residual chemical powder. The bin shall be constructed of stainless steel Type 316S13 to BS EN 10029, BS EN 10259, BS EN 10258, BS EN 10048, BS EN 10095, BS EN 10051 or other suitable corrosion and abrasive resistant suitable material approved by the S.O..

(x) Conveyors General Requirements

1) The General requirement for the screw and belt conveyor shall be in accordance with the General Specification G4.6 Conveyors and specific requirement for each individual chemical shall be as specified;

2) Screw conveyors shall be totally enclosed in a fabricated steel 'U' trough with removable top cover and Perspex inspection panel. Where required for outdoor installation, the conveyors shall be fully weatherproofed. They shall comply with BS 4409 Part 1;

3) The filling of the trough shall be limited to a maximum of 30% and the angle of inclination of the conveyor shall generally not exceed 15 degrees to the horizontal. For inclination angles more than 30 degrees, fully lined screw conveyor requirements shall be required as stated in General Specification G.4.6 Conveyors;

4) Screw blades shall be of the continuous helicoidal type with mild steel flights welded to a central shaft. The pitch of the screw shall be uniform or graduated to suit the application;

5) The central shaft shall be heavy duty steel tube having sufficient strength in bending and tension to eliminate the need for intermediate support by hanger bearings;

6) End bearings shall be provided for rigidly supporting the flighted shaft and due allowance made for thermal expansion and contraction;

7) Drive motors shall be geared and provided with a chain and sprocket drive arranged within a guard. The motor shall be preferably mounted on top;

8) When used for metering purposes, screw conveyors shall be controlled by means of a timer. The reproducibility of metering conveyors shall be better than ± 5% by weight; and

9) Subject to the S.O.'s approval, the manufacturer's protective coating system shall be as for silos.

(xi) Horizontal Screw Conveyors

1) Horizontal screw conveyor shall be of trough type. Cross connections of horizontal and vertical screw conveyors for the formation of a chemical transfer system may be formed from tube type shell. The drive to the screw conveyor shall be by motor directly coupled to the drive-end gudgeon and bolted to a supporting stool forming the end plate of the conveyor. The flight of the screw conveyor shall be supported by sealed-for-life bearings. The motor shall have a degree of protection of IP 55 to BS EN 60034-5;

2) Conveyor trough shall be covered by lids with lapped edges for stiffness. Lids shall seat on rubber or other suitable material to completely seal off against the escape of dust and shall be secured by toggle catches of heavy industrial pattern. All joints shall be flanged and gasketed to prevent the escape of chemical dust;

3) Intermediate bearings shall be self-lubricating, carried on adjustable hangers, and capable of easy exchange of bushes. The spacing of intermediate bearings shall not exceed 2.5 m;

4) The sizing of the screw conveyor shall be such that when handling the rated capacity of the equipment discharging into the screw conveyor, the contents shall not flood the hanger bearing; and

5) The flight of the screw conveyor shall be at least 4 mm thick fabricated from stainless steel strip and continuously welded to the shaft to form a varying pitch helicoid flight. Where outlets occur at the end of conveyors, the end flight shall be opposite-handed to prevent material packing into the end plate.

(e) Vertical / Inclined Screw conveyor

1) Vertical/inclined screw conveyor shall be of tube type complete with inspection openings at cross connections. The drive to the screw conveyor shall be by motor directly coupled to the drive-end gudgeon. The drive shall be either bolted to a supporting stool forming the end plate or flange mounted to the conveyor. The flight of the screw conveyor shall be supported by sealed-for-life bearings. The motor shall have a degree of protection of IP 55 to BS EN 60034-5;

2) The conveyor tube shall be formed by continuously spiral welded stainless-steel plate of not less than 5 mm thick and stainless-steel inspection panels shall be at least 2 mm thick with lapped edges for stiffness. Inspection panels shall seat on rubber or other suitable material to completely seal off against the escape of dust and shall be secured by toggle catches of heavy industrial pattern;

3) The flight of the screw conveyor shall be supported from intermediate bearings of the sealed-for-life design. Design of which the weight of the flight is supported by the drive motor shall not be accepted. Purposely made brackets shall be provided and installed for the support of the screw conveyor. Arrangement shall be made to provide access for routine inspection, maintenance and removal of the flight of the screw conveyor;

4) The vertical/inclined screw conveyor shall be designed for transferring chemical powder from a low level to a higher level. Powdered chemical entering the receiving port shall be lifted by continuous rotation of the flight whereas the dropping shall be achieved by gravity. Chemicals shall be discharged at the outlet chute of the screw conveyor in the form of a discharging duct at an inclined angle to the longitudinal axis of the conveyor. All joints shall be flanged and gasketed to prevent the escape of dust;

5) The direction of rotation of the screw conveyor shall be designed to suit the associated upstream screw conveyor such that powdered chemical shall be discharged at the side port and removed by the screw at the cross connection;

6) The flight of the conveyor shall be at least 4 mm thick fabricated from stainless steel strip and continuously welded to the shaft to form a varying pitch helicoid flight; and

7) Close tolerance fit between screw flight and conveyor casing shall be adopted for the conveyor design to allow efficient feed of powdered chemical with minimum operating noise. To facilitate maintenance, seals and bearings shall be of easily accessible design.

G4.47.9 Local Control Panel and Instrumentations

(a) Unless otherwise specified, dedicated control panels shall be provided for the dosing system. All power and control wiring within the system shall be wired to a stainless-steel marshalling electrical panel or terminal box with appropriate rating to facilitate a single electrical power connection and a minimum number of control wiring connections.

(b) For all chemical dosing system, magnetic flow meter should be provided, for precise flow measurement. The suction of the pump shall be fitted with graduated calibration cylinders, to allow chemical flow drop test measurement, with proper isolation valves.

(c) Weight Element

(i) A separate load cell shall be fitted to each leg, and the readings averaged to eliminate the effect of wind loading, unless otherwise specified;

(ii) Weighing system for each hopper shall consist of four weigh elements and a weight indicating transmitter, along with a panel mounted digital loop powered indicator;

(iii) Local and remote digital indicators shall be provided;

(iv) Dummy cells shall be fitted during erection and replaced by functional cells at the time of final commissioning;

(v) The weigh elements shall be designed and installed so that the elements can be maintained or replaced;

(vi) The weigh element sensors shall produce an output voltage, which is proportional to the weight. The weigh elements shall provide full temperature compensation and have an accuracy of 0.25 percent of rated capacity with a repeatability of 0.10 percent of rated capacity;

(vii) The weight indicating transmitter shall be housed in an IP 66 enclosure, Type 316 stainless steel;

(viii) The weight indicating transmitter shall have a digital display that indicates net, gross, and tare weight. The display shall have six digits, 10 mm characters with the range of 0 to 999,999 with two fixed zeros;

(ix) Installation and calibration of the Weigh elements and Transmitter will be the responsibility of the Contractor and will require that the installation and calibration of the weighing system be checked out, before start-up, by a factory representative of the weigh element supplier; and

(x) All instrument shall comply with the ICA General Specification.

G4.47.10 Dilution Systems

(a) System Design

(i) The dilution water system shall be designed such that it is impossible for chemical to contaminate the plant water supply system by back flow or other means or to cross contaminate other chemical dosing systems. Non-return valves shall not be considered a sufficient method of back flow protection.

(b) System Components

(i) The system shall incorporate all isolating valves as necessary, check valves on the water inlets, flowmeter and flow switch for water supply. An alarm shall be raised if the water flow rate to any dosing point falls below a preset minimum. In automatic systems, the water supply shall start and stop in proper sequence with the corresponding dosing system.

G4.47.11 Pipework

(a) Pipework materials for chemicals shall be selected with due regard to chemical compatibility, corrosion resistance, location, environmental conditions and operating regimes.

(b) Pipework material requirements for individual chemicals shall be according to the relevant section of this specification.

(c) Thermoplastic pipework may be polyvinylidene difluoride (PVDF), polypropylene (PP-R), PVC/cPVC or uPVC lined with GRP where specified.

(d) The minimum size of all the chemical pipes shall be sufficient for maximum dosage of chemical to prevent potential blockage.

(e) Pipework systems shall be located to avoid temperature extremes, excessive vibration and the likelihood of accidental impact damage, especially in areas where personnel access is permitted. If this is not possible, components shall be suitably protected using thermal insulation, electric surface heating, guarding, etc., as appropriate.

(f) If there is a significant likelihood of surface embrittlement occurring with plastic pipework due to exposure to ultra-violet radiation (e.g. on pipework installed outdoors), temperature and impact by wind-blown object, suitable protection of the pipe shall be provided.

(g) Thermoplastic materials shall not be painted as a means of protection but shall be shielded from direct sunlight and adequately guarded against mechanical damage.

(h) Unless otherwise specified, all external chemical pipework shall be laid in trenches provided with removable covers.

(i) Pipework system shall be designed to safely withstand the maximum system pressure. If necessary, components shall be derated, in accordance with the pipework manufacturer's instructions, to allow for high temperatures, repeated pressure fluctuations and/or the corrosive effects of aggressive dosing chemicals (e.g. concentrated sulphuric acid, sodium hypochlorite etc.).

(j) Where pressure and temperature de-rating precludes the use of thermoplastics, suitably lined reinforced plastics, rubber-lined carbon steel or stainless steel shall be used. Stainless steels shall not be used for chemicals containing chlorides.

(k) The pressure drop along pipework systems shall be kept to a practical minimum and shall be carefully considered during dosing pump selection/system design. Pipework sections shall incorporate as few bends as possible and bends, if necessary, shall be of the long radius type.

(l) The wetted internals of all pipework systems (including gaskets and O-rings) shall be resistant to corrosion/degradation by the dosing chemical.

(m) All pipework and hoses shall be adequately supported throughout their run. Pipes shall be fixed to walls or mounted on channel sections fixed to walls or on cantilevered supports off walls. Hanger type supports shall not be acceptable nor shall pipes be underslung from roofs or cantilevered supports or supported on the underside of pipe supports.

(n) Pipework laid on the floor shall be supported on channel sections or similar off the floor.

(o) Pipework shall be supported in accordance with manufacturers' recommendations and adequate provision shall be made for thermal expansion and contraction. Supports shall not be positioned near fittings which would interfere with the natural movement of the pipes. Supports shall be sufficiently wide to offer adequate bearing surface and shall be installed to offer lateral restraint without restricting axial movement of the pipe.

(p) Valves and other devices shall be supported independently of the pipe.

(q) Fixings of pipes shall by clips (cobra type) or clamps. Where clamps are used on plastic pipes, pipe cushioning shall be used and the pipe shall be mounted on channel sections with a plastic strip in between.

(r) All pipe supports shall be stainless steel SS316L or higher grade materials and be chemically compatible to guarantee against corrosion resistance in the specific environment of installation.

(s) Hoses carrying chemical solutions shall be laid on horizontally mounted trays. Hoses shall be securely fixed to trays by clips or similar.

(t) Hoses and pipes shall be laid in such a way that individual pipe/hoses can be removed without dismantling adjacent pipes/hoses.

(u) The racks or trays used to carry chemical pipes/hoses and water service pipes shall not be used for conveying electrical/instrument cable. Where electric/instrument cables are laid in the same trench or gallery with chemical/water pipes, the electric/instrument cables shall be laid on uppermost trays.

(v) Attention shall be paid to the layout of the chemical pipework, which shall be functional and neat in appearance.

(w) All pipework above access ways shall be at not less than 2.5 m above floor level. Pipes laid at or near floor level across access ways shall be suitably protected from damage. Joints in pipelines and fittings shall be sited away from access ways and working areas; where this is unavoidable anti-splash guards over the joints shall be provided.

(x) Pipework connecting to and near pumps shall be rigid and well supported so that strain is not placed on pumps. Suction pipes shall, where possible, be straight and short. Elbows shall be avoided in pipework as far as possible and shall be replaced by 45 degree or long-sweep 90 degree fittings.

(y) Chemical transfer pipework shall be arranged for self-draining either towards the dosing pumps or dosing point. Provision shall be made for draining all pipe sections at low points and for flushing the same with clean water and facilities shall be provided for safe disposal of the drainage.

(z) Pipework including the dosing line shall be provided with flushing connections and drains to facilitate cleaning of pipes/hoses and fittings. The discharge point shall be subject to S.O.'s approval.

(aa) It shall be possible to isolate sections of pipework for flushing with minimum of interruption to dosing. Unless otherwise specified, dosing pipes/hoses and associated injection/diffuser lances/quills/fittings shall be provided in duplicate (1 duty, 1 standby).

(bb) Chlorine and ammonia solution lines shall not be laid inside buildings (not designated for these gases). Where chlorine and ammonia solution lines are laid in dosing chambers and other confined spaces which are accessible by personnel, the Contractor shall provide a permanently installed leak detectors for continuous monitoring of leaks and mechanical ventilation systems.

(cc) All nuts, bolts, washers and screws shall be manufactured from stainless steel SS316L or compatible to the chemical.

(dd) Unless otherwise specified, flanges for steel pipework shall be of the weld-neck type, Type 11 to BS EN 1092 with the appropriate pressure designation but not less than PN10 according to the design pressure for the specific applications.

(ee) Unless otherwise specified or approved by S.O., all joints in rigid metallic and plastic pipework systems shall be fully welded or flanged (i.e. not threaded). If threaded connections and fittings are used, the seal shall be obtained via double ferrule, olive or O-ring type connections that can be dismantled and re-made without the need for new parts i.e. without the need for sealing tape.

(ff) Solvent welded joints in rigid plastic pipework systems shall be made strictly in accordance with the pipework system supplier's instructions. The solvent shall be compatible with the dosing chemical; where special solvents are available for use with aggressive dosing chemicals, these shall be used instead of the standard solvent.

(gg) Pipework connections to tanks shall be of the bolted flange type. Stainless steel SS316L backing rings shall be used on plastic flanged connections and shall be resistant to corrosion/degradation by the dosing chemical.

(hh) All uPVC liners shall be 'hi-impact' type to BS 3505 or BS 3506 Class E, with solvent welded joints and fittings complying with the relevant parts of BS 4346. Reinforced plastics shall comply with BS 6464 and shall contain a uPVC liner as specified. The pipework shall be fabricated and assembled such that only the liner comes into contact with the fluid. The piping system shall be designed and constructed in accordance with BS 7159.

(ii) Where flanged joints are required in thermoplastic pipework, full-face plastic flanges, where available, and stainless steel SS316L backing rings shall be used.

(jj) All Flanges shall conform to BS EN 1092 PN10 minimum, unless otherwise specified.

(kk) Joint rings shall be of Viton encapsulated in PTFE.

(ll) Unless specified otherwise, the chemical piping systems shall be designed, fabricated and constructed to provide double containment pipe to contain any chemical leakage from pipe joints or from accidently damaged portion.

(mm) All dosing line shall have flushing and drainage facility. The dosing lances shall be able to be withdrawn under pressure.

(nn) The double containment piping systems shall comply with the following as a minimum unless specified otherwise:

(i) Primary and Secondary containment piping shall be as defined in the Particular Specification;

(ii) All other unlisted components that are intended for use as pressure retaining components shall have sufficient thickness and reinforcement to be able to maintain the same pressure ratings as the secondary containment piping;

(iii) Interstitial supporting devices used to centre and support the primary piping within the secondary containment piping shall be manufactured from PVC spider clips or C-Type, according to ASTM and ANSI. Annular spacing throughout piping system shall allow for leak detection cable installation. Supporting devices shall allow for leak detection cable installation. Supporting devices shall allow continuous drainage in the annular space to the drain points;

(iv) All fitting shall be pre-assembled and pre-tested by the manufacturer; and

(v) At all piping low points, or at a minimum spacing of 75 metres, a drip leg and drain valve with hose connection shall be provided.

(oo) Chemical which contains Gases

(i) For those chemicals which contains gases, the pipework system shall be designed to avoid and airlock in the system and the following shall apply:

1) The calibration columns or standpipes shall be mounted as close to the pump as possible and vented back to the storage tank to allow the bubbles to escape before the chemical enters the pump;

2) Where applicable, suction pipelines shall be installed sloping downward towards the pump suction;

3) Vent to atmospheres shall not be permitted;

4) Spring loaded discharge check valve shall not be used unless approved by S.O.;

5) The sizing for the suction line shall take consideration of applying low flow velocity and minimising turbulence;

6) A vent valve shall be provided in the discharging pipe closed to the pump head and shall vent back to the storage tank. A solenoid valve shall be provided and shall be open when the pump starts and close after a pre-set time. This shall enable to valve off any backpressure when venting the discharge line and to purge any accumulated gas on start up; and

7) The pump shall be mounted close to the tank and below liquid level in the tank.

(ii) Leak Detection System

1) Shall comply to ICA Specification.

(iii) Testing

1) The secondary containment piping shall be pneumatically tested at design pressure or 1.5 times the working pressure for a minimum of 2.5 hours duration;

2) The external joints shall be soaped and visually inspected for leaks. A pressure regulator and safety relief valve shall be used during the pneumatic test to ensure that over-pressurisation cannot occur; and

3) The testing of the primary and secondary containment piping shall be done in strict accordance with the recommendations of the manufacturer.

(iv) Purging

1) The annular space shall be purged of moisture containing air and foreign matters (dirt, mud, oil) by replacing the volume of air with clean, dry nitrogen or minus 30 degrees C dew point air.

G4.47.12 Isolating Valves

(a) General

(i) All isolating valves shall be positioned with due regard to routine access for operational and maintenance purposes. It shall not be necessary to enter any bunded facility to operate isolating valves; and

(ii) Isolating valves shall be compatible generally with the pipework systems in which they are installed.

(b) PVC / cPVC valves

(i) Isolating valves shall be of the full port ball type with PVC/cPVC body, Teflon seat, Viton O-rings. Assembly nuts and bolts shall be stainless steel. Reflux valve shall be ball check type with PVC/cPVC body;

(ii) End connection fittings may be plain sockets for solvent welding or union coupling type or flanged as appropriate. Flanges shall be drilled to the standard matching the pipe selected; and

(iii) Diaphragm ball valves shall be Weir type, PTFE- / PVDF-lined, PVC / PP / PE body, manual operator indicating, rising stem type with handwheel with adjustable travel stop and shall be in accordance with BS EN 13397 or equivalent.

(c) Actuation for Valve

(i) Valves of nominal diameter sizes of 50 mm and larger shall be of the diaphragm actuated, spring loaded, single seat composition disc, packless globe valve type controlled by a suitable solenoid pilot valve;

(ii) Valves of nominal diameter sizes less than 50 mm in size shall be of the direct acting single integral seat, full pipe area, globe type. They shall be of all bronze construction with tight seating, renewable composition discs and shall be provided with a manual operator; and

(iii) Operating solenoid shall have direct current coils and an integral rectifier for use on a 230 V AC supply, unless otherwise specified. The coils shall be encapsulated in epoxy resin.

G4.47.13 Pressure Relief Valves

(a) Pressure relief valves shall be fitted to the delivery side of all dosing pumps. All materials of construction shall be fully compatible with the chemical handled.

(b) The valves shall discharge in a safe manner to a gulley or sump designed for this purpose. The discharge pipes shall not be manifolded.

(c) A means shall be provided of detecting that a relief valve is operating and of raising an alarm.

G4.47.14 Loading Valves

(a) Unless otherwise specified, loading or back-pressure valves shall be fitted where necessary to the delivery side of the dosing pumps. All materials of construction shall be fully compatible with the chemical handled

G4.47.15 Injection Fittings

(a) Where the chemical is to be dosed into an open channel, weir chamber or upstream of a hydraulic jump, it shall be applied using a distributor. The distributor shall be tubular with orifices drilled at even intervals to ensure uniform distribution of the chemical.

(b) Where it is to be dosed into flow in pipelines, it shall be applied using a suitable arrangement of injection tubes designed for the specified duty flow rate.

(i) Pipelines up to 600 mm in diameter

1) A single horizontal injection tube shall be used extending one third of the pipeline diameter into the fluid stream.

(ii) Pipelines up to 1000 mm in diameter

1) Two injection tubes shall be used mutually located at 90^{o} with their axes at 45^{o} to the horizontal and extending 15% of the pipeline diameter into the fluid stream.

(iii) Pipelines up to 2500 mm in diameter

1) Four injection tubes shall be used mutually located at 90^o with their axes at 45^o to the horizontal and extending 15% of the pipeline diameter into the fluid stream.

(c) The nozzle velocity shall exceed 0.75 m/s or one half of the large pipe velocity whichever is greater.

(d) Distributors and injection tubes shall be of sturdy design and adequately supported. They shall withstand the flow of velocity at the point of application and any flow or turbulence induced vibrations.

(e) The materials of construction of the distributors and injection tubes shall be resistant to erosion and chemical attack over the full range of operating conditions.

(f) The distributors and injection fittings shall be supplied complete with the necessary isolating valves and, where applicable, non-return valves.

(g) Where solutions are dosed into static mixers and the injection unit forms an integral part of the static mixer, the injection fitting design shall be as recommended by the static mixer supplier.

G4.47.16 Pulsation Dampeners

(a) Pulsation dampeners shall be fitted where necessary to the delivery side of chemical dosing pumps to attenuate peak pressure drops and to dampen the flow pulsations, unless otherwise specified.

(b) The dampeners shall be of the vertical air vessel type connected in a manner which minimises the possibility of blockage occurring.

(c) An air supply connection shall be provided on the dampeners to facilitate periodic re charging. The supply shall incorporate a filter, pressure regulator and pressure gauge.

G4.47.17 Dosing Skids

(a) The dosing skids shall be provided and shall consist of a factory-assembled and tested skid-mounted system including dosing pumps and associated components and a control panel. The skid shall consist of a rigid frame fabricated from Kevlar coated carbon steel and have a uPVC backboard for mounting pipework and an integral uPVC bund. The skid shall have external lugs for bolting down to a concrete plinth and lifting eyes to facilitate offloading at site. The bund floor shall slope at a gradient of 1:100 towards one corner draining into a bunded area provided with a sump.

(b) Each dosing pump and its associated components handling undiluted chemical shall be enclosed by transparent plastic anti-spray screens with removable panel to permit access for maintenance whilst the adjacent system remains in service and fully screened. Removal of the panel shall require the use of a tool. Where pipework conveying concentrated chemical to one dosing stream passes through the enclosure housing the other dosing stream, there shall be no joints.

(c) Chemical dosing equipment including skids shall be of substantial construction with all parts designed for long life under working conditions including corrosive atmospheres and intermittent or continuous operation.

(d) Skid materials shall be constructed of materials resistant to the pumped solution.

(e) All wearing parts and items requiring adjustment shall be readily accessible.

(f) All parts which are exposed to corrosive conditions shall be made from corrosion-resistant materials or covered with suitable protective coatings.

(g) Each skid shall be constructed such that the bund wall elevation is flush with the skid bottom unless otherwise indicated. The skids shall be mounted on legs that extend down to the finished floor in the bund area.

(h) Dosing pumps shall be mounted on a plinth local to the solution tank and the pumps and the tank shall be surrounded by a bund.

(i) The dosing pump shall be sized to provide sufficient head to the dosing point as specified.

(j) Unless otherwise specified, chemical dosing pumps shall be in accordance with the General Specification G4.4 Pumps and any specific requirement for each individual chemical as specified in this specification.

G4.47.18 Chemical Plant Safety

(a) The chemical plant shall be designed to a high standard of safety and shall comply with all local statutory requirements and where such local standards do not exist, those of any other relevant bodies recognised or applied locally unless otherwise specified.

(b) The chemical storage and handling plant shall conform to the relevant material safety datasheets, technical service notes and literature published by the potential chemical suppliers and plant manufacturers and any other guidelines published by reputable authorities.

(c) The Contractor shall obtain the approval of the chemical supplier for the design of the reception and storage facilities before proceeding with installation. In the case of toxic or hazardous chemicals the Contractor shall afford the suppliers the opportunity of offering advice on safety precautions to be taken on the design layout, operation and maintenance of the plant.

(d) The Contractor shall provide safety equipment like safety showers, eye baths, eye irrigators, etc. as necessary. Safety equipment shall comply with the General Specification on safety equipment.

(e) The Contractor shall provide all necessary warning and safety signs to BS EN ISO 7010 and BS 5499.

(f) Safety signs shall use local or internationally recognised pictorial representations, such as those in BS EN ISO 7010 and BS 5499.

(g) Signs shall be manufactured from durable, non-fading, and weather resistant materials. Materials shall be of durable quality and chemical resistant for their intended environment, and suitable for both indoor and outdoor use. Signs shall be in stove enamelled aluminium with rounded corners and drilled fixing holes as specified.

(h) Chemical suppliers' notices shall be framed with Perspex cover and a suitable backing board and mounted either on a clear wall just inside the access door or adjacent to plant as appropriate.

(i) Drawings of all warning signs and notices shall be submitted to the S.O. for approval.

(j) Transparent spray guards shall be fitted over the storage tank outlet pipework, within the tank bund. The guards shall be in Perspex or similar sheet mounted horizontally from the side walls of the bunds using suitable brackets.

(k) The dosing pumps shall be provided with screens mounted around the periphery of the dosing pump area or light weight removable transparent covers over the pumps.

(l) Perspex covers shall be provided over joints in the pipework or any other point of leakage to protect personnel from possible leaks.

(m) The delivery areas shall be level.

(n) Chemical Safety Data Sheets (SDS) of all concerned chemicals shall be placed in a durable transparent holder fixed at a conspicuous location on each chemical storage tank.

G4.47.19 Lime Plant

(a) General

(i) In addition to the general requirement, this Section describes the components and applicable requirements which, where relevant to the Contract, shall apply to the supply, installation and testing of lime plant.

(b) Lime Silos Capacity, Design and Fabrication

(i) Silos shall be straight vertical cylinders of carbon steel construction with respectively conical hopper bottom sections unless otherwise specified. The silos and associated structure supports etc shall be designed and endorsed by relevant discipline P.E. and submitted to S.O. for approval;

(ii) The silos shall accommodate the specified mass of hydrated lime with a bulk density of approximately 480 kg/m^3;

(iii) Subject to the requirements of installation, the minimum capacity of a single silo shall be 30 tonnes whilst in multiple silo installations, the capacity of each silo shall be at least 25 tonnes;

(iv) Silos shall be designed, constructed and tested in accordance with the Code of Practice for the Design of Silos, Bins and Bunkers published by the British Material Handling Board, latest edition, and with the relevant parts of BS EN 618:2002 and BS EN 620:2002+A1:2010;

(v) Due allowance shall be made for pressure exerted by material in storage and during filling and emptying;

(vi) Wind loading shall be considered whether or not the silos will ultimately be enclosed in a building; and

(vii) Fabrication of silo storage systems shall be carried out by companies who have constructed satisfactory systems for storing the same material in similar quantities in the past. Proof shall be submitted at time of tender that the proposed manufacturer has been engaged in the fabrication of silo systems for at least 10 years and that he has had silos of the type proposed in satisfactory operation for the same application for at least 5 years.

(c) Filling Arrangement

(i) Silos shall be filled from bulk air pressure discharge vehicles unless otherwise specified;

(ii) The filling pipe shall be at least 100 mm NB carbon steel to BS EN 10216 which shall enter a cylindrical silo tangentially near the top or a rectangular silo downwards through the top. The pipe shall be lined with polythene lining unless otherwise specified;

(iii) The pipe shall rise vertically from the point of connection of the bulk delivery vehicle to the top of the silo and then horizontally. Bends shall have a minimum radius of 1000 mm and the number of bends shall be kept to a minimum, preferably not more than two. Long horizontal lengths of pipe shall be avoided and the total length shall not exceed 40 metres; and

(iv) A 100 mm NB "Unicone" tail piece shall be fitted to the lower end of the filling pipe just above which shall be a hand operated isolating valve of the straight-through type.

(d) Air Filter

(i) Air filters shall be designed to remove fine dust from the total volume of air vented when the silo is being filled. Each filter shall be sized for a throughput greater than the air flow rate used in charging the silo for a period not less than the time taken to transfer the contents of one tanker into the silo;

(ii) Filters shall be in a fully weatherproof casing mounted on top of the silo with a facility for automatic cleaning. The controls for this cleaning facility shall be sited at ground level. Swivel lifting davits or like devices shall be provided to facilitate ease of replacement of filters and transportation of parts and components that may be required in the filters replacement; and

(iii) Where silos are in an enclosed area, each silo shall be provided with two fully rated filters with the necessary quantities of replacement components/parts as specified.

(e) Inspection Manhole and Cover

(i) Each silo shall have a manhole for inspection and maintenance purposes and removable grid bars shall be fitted for safety during inspection. The manhole cover shall be hinged and shall effect an air-tight seal when in the closed position.

(f) Pressure Relief Device

(i) A suitable pressure relief device shall be fitted to the top of each silo to relieve pressure should the air filter become partially or totally blocked or if overfilling occurs during a delivery;

(ii) When in the fully open position, at the most restricted part of the opening, the device shall have a cross sectional area greater than that of the filling pipe;

(iii) Facilities shall be included to prevent any part of the device becoming detached; and

(iv) During normal operation, the device shall not allow emission of dust.

(g) Dehumidifier

(i) Each silo shall be equipped with a dehumidifier of the desiccant type designed to ensure a dry ullage space in the silo and prevent condensation of moisture in the silo;

(ii) The dehumidifier shall dry the air supplied to the silo to a dew point 7.5°C below the ambient temperature. The dried air shall be introduced into the silo at a rate greater than the lime withdrawal rate to create a positive pressure in the silo and shall vent excess air through the air filter unit; and

(iii) The desiccant unit shall comprise of an absorption rotor with several segments and shall be so arranged that the air-drying process is continuous whilst exhausted segment(s) is being reactivated/replaced. The desiccant shall be activated alumina or silica gel. The reactivation of the desiccant shall be effected thermally. The reactivation air system shall be separated from the process air system and the air inlets to both the systems shall be fitted with filters.

(h) High Level Indicator

(i) A high-level indicator of the rotating paddle type or similar or better shall be installed in a suitable position which minimises interference from inflow of the material. The associated control panel shall be located adjacent to the silo filling point and shall have the following features: -

1) Isolating switch;

2) On/off switch for paddle drive;

3) Indicator lamp for PADDLE RUNNING;

4) Indicator lamp for SILO FULL;

5) Audible warning bell;

6) Push-button for LAMP TEST; and

7) Instruction notices to driver.

(ii) The panel enclosure shall be protected to BS EN 60529. The power supply shall be 110 V A.C. unless otherwise specified.

(i) Hopper Outlet

(i) The hopper outlet shall be designed for reliable flow of bulk solids based on cohesive arching and wall friction considerations;

(ii) The angle at the bottom of the silo shall not be less than 60° to the horizontal;

(iii) The outlet shall not be less than 200 mm in diameter or 0.04 m^2 square rectangular cross section;

(iv) An isolating valve of the slide gate type shall be fitted to each outlet; and

(v) Where applicable, the hopper section shall be fitted with an arch breaking device which shall be of the aeration or vibratory type or a combination of both the types. Evidence of proven suitability of the proposed system shall be made available before selection is made.

(j) Support Structure

(i) Support structures for steel silos and steel hoppers shall be in structural steel complying with BS EN 10137, BS 7668 and BS EN 10029 and shall be complete with base plates and anchor bolts.

(k) Man Access

(i) The top of the silo shall be designed as a maintenance platform and be provided with kick-plates, hand railing, walkway, ladders shall be provided as specified;

(ii) Unless otherwise specified, where more than one silo is installed, a walkway between adjacent silos shall be provided also complete with kick-plates and handrailing;

(iii) Access to the top of the silo(s) shall be provided by means of stair access unless otherwise specified. Where more than one silo is installed one stair access and one ladder access with safety cage and at least one intermediate platform shall be provided. The ladder shall comply with BS 4211; and

(iv) Where the bottom outlet valve is above normal working height, a further access platform with kick-plates, handrailing and stairway shall be provided.

(l) Equipment Access

(i) Silos shall be provided with an electrically operated wire hoist designed to handle the heaviest item on top of the silo. Where more than one silo is installed, a monorail shall be provided over the silos to enable the hoist to serve all the silos.

(m) Protective Coating

(i) Steel silos shall be blast cleaned and coated externally as set out in the General Specification on painting and protective coatings; and

(ii) Unless otherwise specified, internal surfaces shall remain untreated.

(n) Screw Conveyors

(i) Screw conveyor system shall be provided if required; and

(ii) The system configuration shall be subject to S.O.'s approval.

(o) Slurry Tanks

(i) Tanks for slurry preparation and dilution of lime slurry shall be vertical cylindrical tanks. Slurry tanks shall be fabricated from mild steel with necessary lining or coating where applicable and designed to hold the required volume of hydrated lime slurry at concentrations up to the specified value.

(ii) Each tank shall be provided with the following components and fittings:

1) Bolted dust-tight cover fitted with a manhole and serving as a maintenance platform;

2) Hand railing around periphery of tank cover and access ladder from floor level;

3) Top entry paddle mixer (or any other mixer type subject to S.O.'s approval) of the slow speed type with driving motor and speed reduction gearbox mounted on structural steel supports spanning the diameter of the tank shall be provided;

4) The gearbox output shaft shall be coupled to the mixer shaft by means of a flexible coupling. The upper end of the mixer shaft shall have a combined journal and thrust bearing and the lower end shall have an inert non-metallic bearing;

5) The paddle blades shall be inclined from the vertical plane. Unless otherwise specified, the mixer shaft and paddles shall be in stainless steel to PD 970 or BS EN ISO 9445-2 grade 316S11;

6) Where necessary for maintenance purposes, a steel lifting beam complying with BS 2853 shall be provided for removal of the mixer and its drive assembly;

7) Sight glass type level gauge with isolating valves and flushing connections shall be provided;

8) Conductivity type level probes for automatic slurry batch preparation and tank changeover, where specified;

9) Powder inlet flange to suit charging system and venting connection with dust filter where applicable;

10) Flanged connections drilled to BS EN 1092 PN 10 for water inlet, slurry inlet (where applicable), slurry outlet, overflow and drain;

11) Where specified, additional connections shall be provided for slurry re-circulation. The tank design shall ensure that insoluble impurities are not passed into the outlet system;

12) Strainer shall be provided in the tank outlet;

13) Unless otherwise specified, the tanks shall be mounted on load cells of the mechanical shear beam type to enable lime usage to be recorded and to control lime slurry preparation. Each tank shall be provided with three live load cells protected to IP 67 as minimum. Facilities shall be built in to allow load cells to be easily replaced when necessary;

14) The tank shall be blast cleaned and coated internally (epoxy system) and externally (chlorinated rubber system) as set out in General Specification for painting and protective coatings; and

15) The automatic flushing and drainage system shall be provided for the slurry tank.

(p) Lime Slurry and Lime Sludge Pumps

(i) Lime slurry and lime sludge pumps shall be peristaltic or reciprocating diaphragm type pumps for the advantage of conveying fluids with high tendency to cause chokage in other types of pumps, unless otherwise specified;

(ii) The pump, motor and drive arrangement shall be mounted on a robust combination baseplate. Unless otherwise specified, only one liquid end shall be driven by the pump motor;

(iii) Transfer pumps shall be progressive cavity type with rotors of hardened steel and stators shall be moulded to metal type using natural or synthetic rubber; and

(iv) Pumps shall be located with the tanks inside a bunded area.

(q) Dilution System

(i) Lime shall be diluted prior to injection to improve mixing at the dosing point. Dilution ratio shall be as specified. The dilution water shall have an alkalinity equal to or less than 10 mg/l as $CaCO_3$ and dissolved carbon dioxide less than 5 mg/l;

(ii) The dilution water system shall be designed such that it is impossible for lime to contaminate the drinking water supply system via the service water system by back-flow or other means or to cross-contaminate other chemical dosing systems;

(iii) The system shall incorporate all isolating valves for water and lime, check valves on the water inlets and flowmeter and flow switch for water supply; and

(iv) An alarm shall be raised if the water flow rate to any dosing point falls below a pre-set minimum. In automatic systems, the water supply shall start and stop in proper sequence with the corresponding dosing system.

(r) Slurry Pipework

(i) Slurry tank outlet manifolds, dosing pump suction manifolds and dosing pump delivery manifolds shall be in seamless carbon steel to BS EN 10216;

(ii) Delivery lines to points of application shall be in reinforced flexible hose unless otherwise specified. Connections between manifolds and dosing pumps may also be made with reinforced flexible hose;

(iii) Slurry hoses shall be secured to horizontally laid trays fixed to duct walls or walls of galleries as necessary. Pipe supports shall be close enough to prevent sagging;

(iv) Tees and crosses shall be used in place of elbows where mechanical cleaning may be required;

(v) Rodding points shall be provided opposite the dosing pump suction off-take points;

(vi) Reducers shall have level inverts and branches shall be taken from the top of a pipe. Long vertical rises shall be avoided especially above the discharge valve of a dosing pump;

(vii) Pipework design shall minimise any tendency for settlement of particles. Provision shall be made for flushing all sections with clean water; and

(viii) Wherever possible, the velocity of the slurry in any part of the pipework system shall be maintained above 0.3 m/s over the entire range of operating conditions.

(s) Slurry Isolating Valves

(i) Manual isolating valves for hydrated lime slurry shall be either ball valves of the full-bore type or diaphragm valves of the straight through type;

(ii) Valve bodies shall be of cast iron whilst balls shall be stainless steel. Flexible diaphragms shall be selected for maximum resistance to abrasion;

(iii) Assembly nuts and bolts shall be in stainless steel; and

(iv) End connections shall be flanged and drilled to BS 1092 PN 10 minimum.

(t) Loading Valves

(i) The valves shall be of the tubular pinch type pressurised externally with compressed air or of a non-clogging alternative approved type. Where pinch type or similar is used, a separate non-return valve shall be provided downstream; and

(ii) Loading or back-pressure valves shall be fitted where necessary to the delivery side of hydrated lime dosing pumps.

(u) Injection Fittings

(i) Where hydrated lime slurry is to be dosed into flow in pipelines, a suitable arrangement of injection tubes shall be used which shall be withdrawable from the pipelines whilst the latter are in service.

(v) Pressure Relief Valves

(i) Pressure relief valves shall be fitted to the delivery side of hydrated lime dosing pumps and shall discharge in a safe manner to a gulley or sump designed for this purpose;

(ii) The valves shall be of a non-choking type free of cavities and crevices likely to block with lime; and

(iii) Provision shall be made for operating the valves manually for purposes of flushing.

G4.47.20 Sodium Hypochlorite Plant

(a) General

(i) In addition to the general requirement, this Section describes the components and applicable requirements which, where relevant to the Contract, shall apply to the supply, installation and testing of sodium hypochlorite plant.

(b) Storage tanks

(i) Sodium hypochlorite solution storage tanks shall be of enclosed vertical cylindrical construction, fabricated from uPVC lined glass reinforced plastic complying with BS EN 13923 and BS EN 13121 or alternatively from high density polyethylene;

(ii) The tanks shall be suitable for storing the solution at the concentration specified. Where delivery of sodium hypochlorite by road tanker or pumped from intermediate bulk containers (IBC) is required, the tanks shall be suitable for solution concentrations up to 15% w/w. Where differing concentration value(s) are specified in the Particular Specifications for the specific intended purpose, the particular specifications shall take precedence;

(iii) Tank level gauge shall be provided as magnetic level indicator type. Wet material shall be suitable for the process fluid and for the process conditions (temperature, pressure). The exterior and housing materials shall be suitable for the environment. The gauge shall be provided with flushing and drain points for service with injurious or clogging substances; and

(iv) Connections

1) Where specified, each storage tank shall be equipped with a filling connection to facilitate charging from a road tanker or pumped supply from an IBC;

2) The overflow connection shall incorporate an internal dip tube to function as a gas trap. The external overflow pipe shall include a siphon break;

3) The outlet arrangement shall incorporate double isolating valves, an in-line strainer and a sample valve. The tank connection shall be positioned to prevent entrainment of gas bubbles; and

4) The vent shall discharge at a safe distance from tank roof level and from adjacent buildings.

(c) Pipework

(i) Pipework materials for hypochlorite service shall be selected with due regard to chemical compatibility, location, environmental conditions and operating regime;

(ii) Specific material requirements are set out below for the pipework of sodium hypochlorite storage tanks;

(iii) In the case of pipework located outdoors, consideration shall be given to the effects of ultra-violet rays, temperature and impact by wind-blown objects. Thermoplastic materials shall not be painted as a means of protection but shall be shielded from direct sunlight and adequately guarded against mechanical damage;

(iv) Where pressure and temperature de-rating precludes the use of thermoplastic materials, suitably lined reinforced plastics or rubber lined carbon steel shall be used;

(v) Provision shall be made for flushing all pipe sections with clean water;

(vi) Pipework shall be supported in accordance with manufacturers' recommendations and adequate provision shall be made for thermal expansion and contraction;

(vii) Where Carbon steel pipework are used, it shall be specified by nominal diameters in accordance with BS 3600 Table 1 and shall be in seamless material to BS 3602 grade 360 or 430. Pipework for brine or sodium hypochlorite service shall be lined with natural rubber 3 mm in thickness or alternative material approved by the S.O.;

(viii) Where uPVC pipework is used, it shall be 'hi-impact' type to BS 3505 or BS 3506 Class E with solvent-welded joints and fittings complying with the relevant parts of BS 4346. Where flanged joints are required, full-face uPVC flanges and stainless steel SS316L backing rings shall be used;

(ix) Where reinforced plastics are used, these shall comply with BS 6464 and shall contain a uPVC liner as specified above. The pipework shall be fabricated and assembled such that only the liner comes into contact with the fluid. The piping system shall be designed and constructed in accordance with BS 7159; and

(x) Flanges shall conform to BS EN 1092 with PN10 rating as a minimum.

(d) uPVC Valves

(i) Valves shall be of the diaphragm type with PVC/cPVC bodies, Teflon and EPDM diaphragms. Assembly nuts and bolts shall be stainless steel SS316L. Reflux valve shall be ball check type with PVC/cPVC bodies;

(ii) End connections shall be PN10 rating flanges to BS EN 1092; and

(iii) These valves may not be used for isolating tank outlets and drains.

(e) Rubber Lined Valves

(i) Isolating valves shall be of the diaphragm type to BS EN 13397 with weir bodies in grey cast iron to BS EN 1561 grade 220 (or alternative grade allowed by the design standard) having integral flanged ends, a natural rubber ebonite lining, PTFE diaphragms and top works in standard proprietary materials. Assembly nuts and bolts shall be in stainless steel SS316L. The valves shall be tested in accordance with BS EN 12266-1; and

(ii) Isolating valves for tank outlets shall be of the sealed bonnet type.

(f) Pressure-Relief Valves

(i) Pressure-relief valves shall be fitted to the delivery side of sodium hypochlorite dosing pumps and, where applicable, to brine pumps. All materials of construction shall be fully compatible with the fluid handled. Stainless steel shall not be used for internal wetted parts; and

(ii) The valves shall discharge back to the suction side of the pump. Provision shall be made for operating the valves manually for purposes of flushing. The discharge pipes shall not be manifolded.

(g) Loading Valves

(i) Loading or back-pressure valves shall be fitted where necessary to the delivery side of dosing pumps; and

(ii) The valves shall be of the tubular pinch type pressurised externally with compressed air or of an alternative approved type.

(h) Pulsation Dampeners

(i) Pulsation dampeners shall be fitted where necessary to the delivery side of dosing pumps to attenuate peak pressure drops and dampen flow pulsations;

(ii) The dampeners shall be of the vertical air vessel type or alternative approved type; and

(iii) An air supply connection shall be provided on the dampeners to facilitate periodic recharging. The supply shall incorporate a filter, pressure regulator and pressure gauge.

(i) Injection Fittings

(i) Where sodium hypochlorite is to be dosed into an open channel, weir chamber or downstream of a hydraulic jump, it shall be applied using a submerged distributor;

(ii) The distributor shall be tubular with orifices drilled at even intervals to ensure uniform distribution of solution; and

(iii) Where sodium hypochlorite is to be dosed into flow in pipelines, it shall be applied using a suitable arrangement of injection tubes designed for the specified duty flow rates. In situations where scaling may occur at the point of application, the injection tubes shall be withdrawable from the pipelines whilst the latter are in service.

(iv) Pipelines up to 600 mm in diameter

1) A single horizontal injection tube shall be used extending one third of the pipeline diameter into the fluid stream.

(v) Pipelines up to 1000 mm in diameter

1) Two injection tubes shall be used mutually located at 90° with their axes at 45° to the horizontal and extending 15% of the pipeline diameter into the fluid stream.

(vi) Pipelines up to 2500 mm in diameter

1) Four injection tubes shall be used mutually located at 90° with their axes at 45° to the horizontal and extending 15% of the pipeline diameter into the fluid stream.

(j) The nozzle velocity shall exceed 0.75 m/s or one half the large pipe velocity whichever is greater.

(k) Distributors and injection tubes shall be of sturdy design and adequately supported. They shall withstand the flow velocity at the point of application and any flow or turbulence-induced vibrations.

(l) The materials of construction of the distributors and injection tubes shall be resistant to erosion and chemical attack over the full range of operating conditions.

(m) When a static mixer is used for mixing sodium hypochlorite solution the injection fitting shall be designed and installed in accordance with the static mixer supplier's requirements.

(n) The distributors and injection fittings shall be supplied complete with the necessary isolating valves and, where applicable, non-return valves.

G4.47.21 Sodium Hydroxide (Caustic Soda) Plant

(a) General

(i) In addition to the general requirement, this Section describes the components and applicable requirements which, where relevant to the Contract, shall apply to the supply, installation and testing of caustic soda plant.

(b) Materials

(i) Aluminium, aluminium alloys, tin, zinc (galvanising), lead, bronze and brass shall not be used on the caustic soda plant. Iron and steel, natural rubbers (up to a temperature of 80°C) and thermoplastic materials shall be used as specified. PVDF and nylon shall not be used. GRP shall only be used with a PVC lining.

(c) Solution Tanks

(i) Capacity, Design and Fabrication

1) The tank shall be fabricated from FRP or equivalent material, which shall comply with BS 476: Part 6 or equivalent with fire propagation index meeting latest FSSD requirements, to give maximum resistance to aqueous solution of sodium hydroxide. The tank construction shall be robust and suitable for installation on a concrete plinth. The solution tank shall be suitable for installation inside a compartment with a bund wall for retaining sodium hydroxide solution in case of tank leakage and shall be suitable for the rise in temperature due to heat evolved during dilution. The dilution system shall be designed to minimise local boiling and shall be provided with mixing facilities. In the dilution process, caustic soda shall be added to the water. Unless otherwise specified, the tanks shall be located within a building;

2) Alternatively, tanks (up to 50% w/w NaOH) can be enclosed vertical cylindrical construction, fabricated from steel to BS EN 14015 or uPVC lined glass reinforced plastic complying with BS EN 13923 and BS EN 13121-3; and

3) Mixing of contents during preparation shall be by a mixer.

(ii) Tank Heating

1) Unless otherwise specified, a tank heating arrangement will be required only if the ambient temperature falls below 15 ^{0}C; and

2) The materials of construction shall be suitable for receipt of caustic soda deliveries at temperatures up to 55 ^{0}C.

(iii) Filling Arrangement

1) The filling pipe shall be carbon steel or uPVC at the appropriate pressure rating and wall thickness schedule.

(iv) Outlet Arrangement

1) The tanks shall be provided with an internal plug valve for sealing the outlet in an emergency. If it is not practicable to fit the internal plug valve, two isolating valves shall be installed in series directly at the tank outlet with one of these valves operable from outside the bund.

(v) Temperature Sensor

1) A temperature sensor shall be installed for local and remote indication together with an alarm set at temperature slightly higher than the maximum allowable temperature at 50 ^{0}C for the sodium hydroxide solution for the protection of FRP tank.

(d) Mixer

(i) Mixer impellers may be of single or dual arrangement. No bottom support bearing is to be incorporated in the tank as stabilizers. The motor and gearbox shall be fixed to a combined baseplate;

(ii) The mixer shaft shall be of adequate stiffness to minimize flexing and so driven that no flexural load from the shaft is transmitted to the gearbox or motor;

(iii) All wetted parts, bedplates and holding down bolts shall be of 316L stainless steel or superior materials suitable for sodium hydroxide solution application; and

(iv) Other components exposed to splashing shall be suitably protected to the S.O.'s approval.

(e) Solution Preparation Tanks

(i) Where solution preparation tanks are required, solution preparation tanks shall be constructed of steel or thermoplastic and shall be suitable for the rise in temperature due to heat evolved during dilution. The dilution system shall be designed to minimise local boiling and shall be provided with mixing facilities. In the dilution process, caustic soda shall be added to the water.

(f) Recirculation Pumps

(i) Recirculation pumps shall be of the centrifugal type with cast iron casing, stainless steel or cast-iron impeller and stainless-steel shaft. Glands packed with mechanical seals shall be used;

(ii) Pump glands shall be covered by an anti-splash guard and the pump shall be housed over a mild steel catchment tray which drains to a sump;

(iii) Alternatively, glandless centrifugal pumps with magnetic couplings shall be used;

(iv) As an alternative to centrifugal pumps, positive displacement pumps of the reciprocating type or progressive cavity type may be used and subject to S.O.'s approval;

(v) Connections shall be provided for draining and flushing out the pumps; and

(vi) Recirculation pumps shall be in bunded areas and shall be fully accessible for operation and maintenance purposes without personnel having to enter the bund itself.

(g) Pipework

(i) Pipework shall be stainless steel Type 316S13 to BS EN 10029, BS EN 10259, BS EN 10258, BS EN 10048, BS EN 10095, BS EN 10051. Plastic pipework shall only be used for diluted caustic soda;

(ii) Stainless steel pipework shall be type 316S13 to BS EN 10216 and BS EN 10217 and shall be flanged. Fittings shall comply with BS EN 10253 Part 4. All welding shall be carried out to BS 4677 and procedures shall be subject to approval in accordance with BS EN ISO 15614-1:2004+A2:2012. Welders shall be tested by an independent testing authority and shall satisfy the requirements of BS EN ISO 9606-1;

(iii) All welds shall be dressed;

(iv) Alternative plastic materials such as polypropylene (PP) may be used in certain circumstances subject to approval by the S.O.;

(v) Flanges shall conform to BS EN 1092 with minimum PN 10 where appropriate; and

(vi) Joint rings for plastic pipework shall be of Viton encapsulated in PTFE.

(h) Isolating Valves

(i) Isolating valves shall be of the diaphragm type with uPVC bodies, glass filled polypropylene top works and PTFE diaphragms. Assembly nuts and bolts shall be stainless steel;

(ii) End connection fittings may be plain sockets for solvent welding or union coupling type or flanged as appropriate. Flanges shall be to BS EN 1092 with minimum PN16 or PN 25 where appropriate;

(iii) These valves may not be used for isolating tank outlets and drains unless otherwise approved by the S.O.;

(iv) Main isolating valves for tank outlet and drain shall be of the tapered plug type with PTFE sleeve and cast-iron body; and

(v) All valves shall be tested in accordance with BS 12266: Part 1.

(i) Rubber Lined Valves

(i) General isolating valves requiring rubber-lining shall be of the diaphragm type to BS EN 13397 with weir bodies in grey cast iron to BS EN 1561 grade 220 (or alternative grade allowed by the design standard) having integral flanged ends, a natural rubber ebonite lining, PTFE diaphragms and top works in standard proprietary materials; and

(ii) Assembly nuts and bolts shall be in stainless steel. The valves shall be flanged to BS EN 1092 with minimum PN 10 where appropriate.

G4.47.22 Sulphuric Acid Plant

(a) General

(i) In addition to the general requirement, this Section describes the components and applicable requirements which, where relevant to the Contract, shall apply to the supply, installation and testing of sulphuric acid plant.

(b) Storage Tanks

(i) Capacity, design and fabrication;

(ii) Storage tanks for 50 % w/w H2SO4 shall be of enclosed vertical or horizontal cylindrical construction, fabricated in either lined glass reinforced plastic or thermoplastic materials. Where differing concentration value(s) are specified in the Particular Specifications for the specific intended purpose, the particular specifications shall take precedence;

(iii) Glass reinforced plastic tanks shall be as specified above; and

(iv) Thermoplastic tanks shall be as specified above.

(v) Steel Tanks

1) Steel tanks shall comply with BS EN 14015. The tanks shall be rubber lined internally to BS 6374.

(vi) Filling Arrangement

1) The filling point isolating valves shall be 50 mm nominal diameter flanged PN10 sleeved plug valves lined with PTFE throughout and fitted with a key-operated mechanical interlock which will release the key when in the fully closed position and otherwise hold the key captive. The interlock shall accept the key from the interlock serving the road vehicle delivery bund drain valve. The filling line drain valves shall be of the same valve type of 25 mm nominal diameter. Where stainless steel pipework is used, then both plug valves shall also be in stainless steel.

(c) Dilution Systems

(i) Sulphuric acid solution shall be diluted prior to injection to improve mixing. The dilution ratio shall generally be as specified. Where necessary, dilution tees with PTFE coating may be implemented.

(d) System Design

(i) The dilution water system shall be designed such that it is impossible for sulphuric acid to contaminate the drinking water supply system by back-flow or in any other way.

(e) System Components

(i) The system shall incorporate all isolating valves for water and the chemical, check valves on the water inlets, flowmeter, flow regulating valves and flow switch for water supply.

(ii) An alarm shall be raised if the water flow rate to any dosing point falls below a preset minimum.

(iii) In automatic systems, the water supply shall start and stop in proper sequence with the corresponding dosing system.

(f) Pipework

(i) Carbon steel pipework compatible only for certain concentrations of sulphuric acid shall be in accordance with the requirements below and shall be lined with natural rubber 3 mm thick to BS 6374;

(ii) Carbon steel pipework shall be specified in nominal diameters in accordance with BS 3600 Table 1 and shall be in seamless material to BS EN 10216 and BS EN 10217 grade 360 or 430;

(iii) Carbon steel fittings shall comply with BS EN 10253;

(iv) Unless otherwise specified, flanges for steel pipework shall be of the weld neck type, Type 11 to BS EN 1092 with the appropriate pressure designation but not less than PN10. Valves and other devices mounted in the pipework shall be supported independently of the pipes to which they connect;

(v) Welding of carbon steel shall be carried out in accordance with BS 2633 Class 1 arc welding. All welders shall be tested by an independent inspection authority and shall satisfy the requirements of BS EN 287–1; and

(vi) All fabrication and corrosion protection such as lining of carbon steel shall be carried out at the manufacturers' works. Coating shall comply with the General Specification on painting and protective coatings.

(g) Isolating Valves

(i) Isolating valves for the storage tank filling line, outlet and drain shall be sleeved plug valves as set out above except that the bodies shall be lined with PTFE throughout;

(ii) Other isolating valves shall also meet this specification in instances where the pressure rating exceeds PN10;

(iii) All other isolating valves shall be of the diaphragm type to BS EN 13397 with weir bodies in grey cast iron to BS EN 1561 grade 220 (or alternative grade allowed by the design standard) or alternatively carbon steel, a natural rubber ebonite lining, PTFE diaphragm end top works in proprietary materials;

(iv) Assembly nuts and bolts shall be in stainless steel. The valves shall have integral flanges to BS EN 1092 PN10; and

(v) All valves shall be tested in accordance with BS EN 12266-1.

G4.47.23 Ferric Chloride Plant

(a) General

(i) In addition to the general requirement, this Section describes the components and applicable requirements which, where relevant to the Contract, shall apply to the supply, installation and testing of ferric chloride plant.

(b) Storage Tanks

(i) Capacity, design and fabrication;

(ii) Storage tanks for 41% w/w Ferric Chloride shall be of enclosed vertical cylindrical construction, fabricated in either lined glass reinforced plastic or thermoplastic materials. Where differing concentration value(s) are specified in the Particular Specifications for the specific intended purpose, the particular specifications shall take precedence;

(iii) Glass reinforced plastic tanks shall be as specified above; and

(iv) Thermoplastic tanks shall be as specified above.

(v) Steel tanks

1) Steel tanks shall comply with BS EN 14015. The tanks shall be rubber lined internally to BS 6374.

(vi) Filling Arrangement

1) The pipe shall be in carbon steel to BS EN 10216 and BS EN 10217 with 3 mm thick natural rubber to BS 6374 or polypropylene lining and shall be used as specified herein; and

2) The filling point isolating valves shall be 50 mm nominal diameter flanged PN10 sleeved plug valves lined with PTFE throughout and fitted with a key-operated mechanical interlock which will release the key when in the fully closed position and otherwise hold the key captive. The interlock shall accept the key from the interlock serving the road vehicle delivery bund drain valve. The filling line drain valves shall be of the same valve type of 25 mm nominal diameter. Where stainless steel pipework is used, then both diaphragm valves shall also be in stainless steel.

(c) Dilution Systems

(i) Ferric chloride solution shall be diluted prior to injection to improve mixing. The dilution ratio shall generally be as specified. Where necessary, dilution tees with PTFE coating may be implemented.

(d) Pipework

(i) Carbon steel pipework compatible only for certain concentrations of ferric chloride shall be in accordance with the requirements below and shall be lined with natural rubber 3 mm thick to BS 6374;

(ii) Carbon steel pipework shall be specified in nominal diameters in accordance with BS 3600 Table 1 and shall be in seamless material to BS EN 10216 and BS EN 10217 grade 360 or 430;

(iii) Carbon steel fittings shall comply with BS EN 10253;

(iv) Unless otherwise specified, flanges for steel pipework shall be of the weld neck type, Type 11 to BS EN 1092 with the appropriate pressure designation but not less than PN10. Valves and other devices mounted in the pipework shall be supported independently of the pipes to which they connect;

(v) Welding of carbon steel shall be carried out in accordance with BS 2633 Class 1 arc welding. All welders shall be tested by an independent inspection authority and shall satisfy the requirements of BS EN 287–1; and

(vi) All fabrication and corrosion protection such as lining of carbon steel shall be carried out at the manufacturers' works. Coating shall comply with the General Specification on painting and protective coatings.

(e) Isolating Valves

(i) Isolating valves for the storage tank filling line, outlet and drain shall be sleeved plug valves as set out above except that the bodies shall be lined with PTFE throughout;

(ii) Other isolating valves shall also meet this specification in instances where the pressure rating exceeds PN10;

(iii) All other isolating valves shall be of the diaphragm type to BS EN 13397 with weir bodies in grey cast iron to BS EN 1561 grade 220 (or alternative grade allowed by the design standard) or alternatively carbon steel, a natural rubber ebonite lining, PTFE diaphragm end top works in proprietary materials;

(iv) Assembly nuts and bolts shall be in stainless steel. The valves shall have integral flanges to BS EN 1092 PN10; and

(v) All valves shall be tested in accordance with BS EN 12266-1.

G4.47.24 Sodium Silicofluoride Plant

(a) General

(i) In addition to general requirement as above, this Section contains requirements which, where relevant to the Contract, shall apply to the supply, installation and testing of sodium silicofluoride plant.

(b) Unloading and Storage Facilities

(i) Means shall be provided for unloading sodium silicofluoride from supplier's delivery vehicles and transferring the same to store. The packages may be in the form of palletised or individual bags;

(ii) Manual handling of packages within the store using sack trolleys or similar will be acceptable only when the maximum consumption does not exceed 100 kg/day or as otherwise specified. Any lifting of packages above floor levels shall be fully mechanised;

(iii) The material shall be stored in their original packages in a cool and dry area on pallets; and

(iv) All necessary steps shall be taken to alleviate hazards from sodium silicofluoride dust and to contain spillages during handling operations.

(c) Solution Concentrations

(i) Stock solutions of sodium silicofluoride shall be prepared at a concentration less than its solubility at the lowest temperature of the water used for dissolution;

(ii) The solution preparation system shall be designed to accurately measure the quantity of the fluoride and the volume of water used to prepare a batch of solution; and

(iii) All areas where fluoride dust is likely to be generated shall be provided with mechanical dust extraction/ filtration systems.

(d) Bag Dump System (Charging System)

(i) The bag dump system (charging system) shall be provided with a bag loader for manually charging the hopper which shall consist a hinged lid to place the bag and a spike to retain the bag when the lid is in the closed position. The storage hopper shall be provided with a dust extractor complete with a filter;

(ii) Any lifting equipment such as hoists necessary to raise chemical bags from storage area floor level to tank charging level shall be provided;

(iii) All areas where fluoride dust is likely to be generated shall be provided with ambient air mechanical dust extraction/ filtration systems; and

(iv) Detailed requirements are specified in the section on Solid Chemical Unloading and Storage.

(e) Dissolving Tanks

(i) Dissolving tanks shall operate in a rotational batch basis with a cycle frequency as specified;

(ii) The tanks may be constructed from rubber lined mild steel or fibre reinforced polyester or alternative material approved by the S.O.;

(iii) The tanks shall be provided with a bag loader which shall consist of a hinged lid to place the bag, a spike to retain the bag when the lid is in the closed position and coarse screen and shall be connected to a dust extractor;

(iv) Any associated and necessary lifting equipment to raise chemical bags from storage area floor level to tank charging level shall be provided;

(v) Unless specified, each tank shall be provided with a turbine type mixer constructed of stainless steel to PD 970 or BS EN ISO 18286 grade 316S11. Circular tanks shall be fitted with baffles to minimise vortex formation;

(vi) The following pipe connections shall be provided;

1) Water inlet;

2) Solution outlet;

3) Overflow;

4) Drain; and

5) Level gauge.

(vii) The tank outlet shall be provided with a strainer; and

(viii) Where applicable, suitable level switches or probes shall be provided for operational purposes including low level, high level and overflow level.

(f) Dilution Systems

(i) Sodium silicofluoride solution shall be diluted prior to injection to improve mixing. The dilution ratio shall generally be greater than 10.

(g) Dosing Pumps

(i) Dosing pumps shall be mounted on a plinth local to the solution tank and the pumps and the tank shall be surrounded by a bund;

(ii) The pumps shall be of the metering or reciprocating diaphragm type driven by electric motors or as specified;

(iii) Unless otherwise specified, only one liquid end shall be driven by the pump motor. The pump, motor and drive arrangement shall be mounted on a robust combination baseplate; and

(iv) The pump heads and diaphragms shall be manufactured from thermoplastic materials suitable for the duty conditions.

(h) Pipework

(i) Stainless steel pipework shall be type 316S13 to BS EN 10216 . Pipe couplings shall be light series compression Type A to BS EN 1452;

(ii) uPVC pipework shall be `hi impact' type to BS 3505 or BS 3506 Class E with solvent welded joints and fittings complying with the relevant parts of BS EN 1481 and BS EN 1452. Where flanged joints are required, full face uPVC flanges and stainless steel SS316L steel backing rings shall be used;

(iii) Where reinforced plastics pipe are used, these shall comply with BS 6464 and shall contain a uPVC liner as specified above for uPVC pipework. The pipework shall be fabricated and assembled such that only the liner comes into contact with the fluid. The piping system shall be designed and constructed in accordance with BS 7159;

(iv) Alternative plastic materials such as polypropylene and polyvinylidene difluoride (PVDF) may be used in certain circumstances subject to approval by the S.O.;

(v) Joint rings for steel pipework shall be compressed fibre or proprietary alternative approved by the S.O.; and

(vi) Pipework shall be double contained unless it in run along trench or over a containment bund or instructed by S.O..

(i) Isolating Valves

(i) uPVC Valves

1) Isolating valves shall be of the diaphragm type with uPVC bodies, glass filled polypropylene top works and PTFE diaphragms. Assembly nuts and bolts shall be stainless steel SS316L;

2) End connection fittings may be plain sockets for solvent welding or union coupling type or flanged as appropriate. Flanges shall be to BS EN 1092 PN10 minimum; and

3) These valves may not be used for isolating tank outlets and drains unless otherwise approved by the S.O..

(ii) Stainless Steel Valves

1) Isolating valves shall be of the diaphragm type to BS EN 13397 with weir bodies in stainless steel 316C16 to BS EN 10293, having integral flanged ends, PTFE diaphragms and top works in standard proprietary materials. Assembly nuts and bolts shall be in stainless steel SS316L. The valves shall be flanged to BS EN 1092 PN10 minimum;

2) Isolating valves for tank outlets shall be of the sealed bonnet type; and

3) All valves shall be tested in accordance with BS EN 12266-1.

G4.47.25 Ammonium Sulphate Plant

(a) General

(i) In additional to the general requirement as above, this Section contains requirements which, where relevant to the Contract, shall apply to the supply, installation and testing of an ammonium sulphate plant.

(b) Storage Tanks

(i) Storage tanks for ammonium sulphate (e.g. 25% w/w) shall be of enclosed vertical cylindrical construction, fabricated from fibre glass reinforced polyester or uPVC lined glass reinforced plastic complying with BS EN 13923 and BS EN 13121 or alternatively from high density polyethylene. Where differing concentration value(s) are specified in the Particular Specifications for the specific intended purpose, the particular specifications shall take precedence.

(c) Pumps

(i) Variable stroke mechanisms shall be incorporated in the drive arrangement to allow infinitely variable adjustment of pump output by means of a micrometer, hand wheel or similar mechanical device whilst the pump is running;

(ii) Where the pump is part of an automatic coagulation control or other process control loop, the stroke mechanism shall be fitted with a three-phase bi directional motor with torque limiter and automatic stops at both extremes of travel. A position feedback potentiometer shall be provided to facilitate control and remote indication of position. The operational range of stroke adjustment shall be not less than 6:1;

(iii) Where the pump output is specified to be proportional to flow past the dosing point, then the pump output shall be varied by altering the pump speed and not by altering the pump stroke;

(iv) The pump head shall be driven through a totally enclosed speed reduction gearbox with integral reciprocating drive device of the adjustable crank or mechanical lost motion type. The gearbox and reciprocating drive shall be oil bath lubricated. The unit shall incorporate filling and drain plugs for oil and an oil level indicator;

(v) Drive motors shall be of the three phase cage induction type whether for fixed speed or variable speed operation;

(vi) Where variable speed operation is specified, the speed turn down ratio shall be not less than 5:1;

(vii) Transfer pumps shall be progressive cavity type with rotors of hardened steel and stators shall be moulded to metal type using natural or synthetic rubber;

(viii) For horizontal end suction centrifugal dosing, it shall be back pull out type and shall be used for dosing to final water. Pumps shall be constructed in suitable plastic materials such as glass filled epoxy resin or high density polyethylene with external metal armouring; and

(ix) The pumps shall be of the mechanical seal type. Alternatively, sealless magnetic drive pumps shall be used.

(d) Pipework

(i) Stainless steel pipework shall be type 316S13 to BS EN 10216 Part 5 with tolerances in outside diameter to BS ISO 8434-2, BS EN ISO 8434-1 and tested in accordance with Category 2 requirements. Pipe couplings shall be light series compression to BS ISO 8434-2;

(ii) uPVC pipework shall be as specified above;

(iii) Reinforced plastics shall be as specified above;

(iv) Alternative plastic materials such as polypropylene (PP-R) and polyvinylidene difluoride (PVDF) may be used in certain circumstances subject to approval by the S.O.; and

(v) Joint rings for steel pipework shall be proprietary and approved by the S.O..

(e) uPVC Valves

(i) Isolating valves shall be of the diaphragm type with uPVC bodies, glass filled polypropylene top works and PTFE diaphragms. Assembly nuts and bolts shall be stainless steel SS316L;

(ii) End connection fittings may be plain sockets for solvent welding or union coupling type or flanged as appropriate. Flanges shall be to BS EN 1092 with minimum PN16 or PN 25 where appropriate; and

(iii) These valves may not be used for isolating tank outlets and drains unless otherwise approved by the S.O..

(f) Stainless Steel Valves

(i) Isolating valves shall be of the diaphragm type to BS EN 13397 with weir bodies in stainless steel 316C16 to BS EN 10293, having integral flanged ends, PTFE diaphragms and top works in standard proprietary materials. Assembly nuts and bolts shall be in stainless steel SS316L. The valves shall be flanged to BS EN 1092 with minimum PN10 or PN 16 where appropriate;

(ii) Isolating valves for tank outlets shall be of the sealed bonnet type; and

(iii) All valves shall be tested in accordance with BS EN 12266-1, BS EN 12266-2.

G4.47.26 Citric Acid Plant

(a) General

(i) In additional to the general requirement as above, this Section contains requirements which, where relevant to the Contract, shall apply to the supply, installation and testing of citric acid plant.

(b) Storage Tanks (for dilute acid)

(i) Storage tanks for citric acid (e.g. 50% w/w) shall be of enclosed vertical cylindrical construction, fabricated from fibre glass reinforced polyester or uPVC lined glass reinforced plastic complying with BS EN 13923 AND BS EN 13121 or alternatively from high density polyethylene. Where differing concentration value(s) are specified in the Particular Specifications for the specific intended purpose, the particular specifications shall take precedence.

(ii) Tank level gauge shall be provided as magnetic level indicator type. Wet material shall be suitable for the process fluid and for the process conditions (temperature, pressure). The exterior and housing materials shall be suitable for the environment. The gauge shal be provided with flushing and drain points for service with injurious or clogging substances.

(c) Pipework

(i) Stainless steel pipework shall be type 316S11 to BS EN 10216 and BS EN 10217. Flanges shall be of stainless steel type 316C12 to BS EN 10213 if cast type or type 316S11 to BS EN 10222 if forged;

(ii) Thermoplastic pipework may be uPVC, polyvinylidene difluoride (PVDF), polypropylene or uPVC lined GRP or fibre reinforced polyester;

(iii) All uPVC pipework used as liners shall be `hi-impact` type to BS EN 1452 Parts 1 to 5 or BS 3506 Class E, with solvent welded joints and fittings complying with the relevant parts of BS EN 1452. Reinforced plastics shall comply with BS 6464 and shall contain an uPVC liner as specified. The pipework shall be fabricated and assembled such that only the liner comes into contact with the fluid. The piping system shall be designed and constructed in accordance with BS 7159;

(iv) In the case of pipework located outdoors, consideration shall be given to the effects of ultra violet rays, temperature and impact by wind blown objects. Thermoplastic materials shall not be painted as a means of protection but shall be shielded from direct sunlight and adequately guarded against mechanical damage;

(v) Where pressure and temperature de rating precludes the use of thermoplastic materials, suitably lined reinforced plastics or rubber lined carbon steel shall be used;

(vi) Provision shall be made for flushing all pipe sections with clean water;

(vii) Pipework shall be supported in accordance with manufacturers' recommendations and adequate provision shall be made for thermal expansion and contraction;

(viii) Where flanged joints are required in thermoplastic pipework, full face plastic flanges, where available, and stainless steel SS316L backing rings shall be used. Flanges shall conform to BS EN1092-1 PN10 minimum;

(ix) Joint rings shall be of Viton encapsulated in PTFE;

(x) Where reinforced plastics such as fibre reinforced polyester or GRP are used, these shall comply with BS6464 and FRP shall contain an uPVC liner if required, as specified above for uPVC pipework. The pipework shall be fabricated and assembled such that only the liner comes into contact with the fluid;

(xi) The piping system shall be designed and constructed in accordance with BS 7159; and

(xii) Flanges shall conform to BS EN1092-1 PN 10 minimum.

(d) Valves

(i) Isolating Valves

1) Isolating valves for the storage tank filling line, outlet and drain and those where the pressure rating exceeds PN10 shall be sleeved plug valves having forged bodies in carbon steel lined with PTFE throughout and with integral flanges, tapered plug with PTFE sleeve, top seal arrangement with PTFE diaphragm and delta seal ring, back up metal diaphragm, floating thrust collar, electrostatic eliminator, four bolt cover and three point self-aligning plug adjustment. The valve design shall ensure that excess pressure in the plug and body cavity of the closed valve is relieved spontaneously towards the direction of high pressure. The valve shall be operated by a forged steel wrench with padlocking facility in the closed position or otherwise by power actuator as specified. Where necessary, a remote operating linkage shall be provided;

2) Valves shall be flanged to BS EN 1092 PN10 minimum;

3) All valves shall be tested in accordance with BS EN 12266-1;

4) All other isolating valves shall be of the diaphragm type to BS EN 13397 with weir bodies in grey cast iron to BS EN 1561 grade 220 (or alternative grade allowed by the design standard) or alternatively carbon steel, a natural rubber ebonite lining, PTFE diaphragm end top works in proprietary materials. Assembly nuts and bolts shall be in stainless steel. The valves shall have integral flanges to BS EN 1092 PN10; and

5) All valves shall be tested in accordance with BS EN 12266-1.

G4.47.27 Sodium Bisulphite Plant

(a) General

(i) In addition to the general requirement as above, this Section contains requirements which, where relevant to the Contract, shall apply to the supply, installation and testing of sodium bisulphite plant.

(b) Storage Tanks

(i) Capacity, design and fabrication;

(ii) Storage tanks for sodium bisulphite (20% w/w to 35% w/w) shall be of enclosed vertical cylindrical construction, fabricated from uPVC lined glass reinforced plastic complying with BS EN 13923 AND BS EN 13121 or from stainless steel type 304S14 to BS EN 10216 and BS EN 10217 or polypropylene or fibre reinforced polyester. The tanks shall be designed for storing the acid at up to 38% w/w concentration. Where differing concentration value(s) are specified in the Particular Specifications for the specific intended purpose, the particular specifications shall take precedence; and

(iii) Filling arrangement

1) Refer to the section on Liquid Chemical Unloading/Filling Station.

(c) Pipework

(i) Stainless steel pipework shall be type 316S11 to BS EN 10216-5 or 10217-7. Flanges shall be of stainless steel type 316C12 to BS EN 10213 if cast type or type 316S11 to BS EN 10222 if forged;

(ii) Thermoplastic pipework may be uPVC, polyvinylidene difluoride (PVDF), polypropylene (PP-R) or uPVC lined GRP or fibre reinforced polyester;

(iii) All uPVC pipework used as liners shall be `hi-impact` type to BS EN 1452 parts 1 to 5 or BS 3506 Class E, with solvent welded joints and fittings complying with the relevant parts of BS EN 1452. Reinforced plastics shall comply with BS 6464 and shall contain an uPVC liner as specified. The pipework shall be fabricated and assembled such that only the liner comes into contact with the fluid. The piping system shall be designed and constructed in accordance with BS 7159;

(iv) Where flanged joints are required in thermoplastic pipework, full face plastic flanges, where available, and stainlesss t e e l SS316L backing rings shall be used. Flanges drilling shall conform to BS EN1092-1 PN10 minimum;

(v) Joint rings shall be of Viton encapsulated in PTFE;

(vi) Where reinforced plastics such as fibre reinforced polyester GRP are used, these shall comply with BS6464 and shall contain an uPVC Liner if required, as specified above for GRP pipework. The pipework shall be fabricated and assembled such that only the liner comes into contact with the fluid;

(vii) The piping system shall be designed and constructed in accordance with BS 7159; and

(viii) Flanges drilling shall conform to BS EN1092-1 PN 10 minimum.

(d) Valves

(i) Isolating valves for the storage tank filling line, outlet and drain and those where the pressure rating exceeds PN10 shall be sleeved plug valves having forged bodies in carbon steel lined with PTFE throughout and with integral flanges, tapered plug with PTFE sleeve, top seal arrangement with PTFE diaphragm and delta seal ring, back up metal diaphragm, floating thrust collar, electrostatic eliminator, four bolt cover and three point self-aligning plug adjustment.

1) The valve design shall ensure that excess pressure in the plug and body cavity of the closed valve is relieved spontaneously towards the direction of high pressure. The valve shall be operated by a forged steel wrench with padlocking facility in the closed position or otherwise by power actuator as specified. Where necessary, a remote operating linkage shall be provided;

2) Valves shall be flanged to BS EN 1092 PN10 minimum;

3) All valves shall be tested in accordance with BS EN 12266-1;

4) All other isolating valves shall be of the diaphragm type to BS EN 13397 with weir bodies in grey cast iron to BS EN 1561 grade 220 (or alternative grade allowed by the design standard) or alternatively carbon steel, a natural rubber ebonite lining, PTFE diaphragm end top works in proprietary materials. Assembly nuts and bolts shall be in stainless steel. The valves shall have integral flanges to BS EN 1092 PN10; and

5) All valves shall be tested in accordance with BS EN 12266-1.

(ii) Loading Valves

1) The valves shall be of the tubular pinch type pressurised externally with compressed air or of an alternative approved type.

G4.47.28 Polyelectrolyte Plant

(a) General

(i) In addition to the general requirement, this Section contains requirements which, where relevant to the Contract, shall apply to the supply, installation and testing of polyelectrolyte plant.

(b) Polyelectrolyte Products

(i) The polyelectrolyte plant shall be suitable for use with the chemical product or selection of products specified in the Particular Specifications.

(c) Polyelectrolyte Plant Configuration

(i) Polyelectrolyte plant for handling powdered or granular material shall consist of all necessary means for charging, feeding, wetting, dissolving, ageing and storage of the specified products;

(ii) The plant shall comprise a single train of the following systems which may be assembled into a packaged unit;

1) Charging, storage, feeding and wetting units;

2) Dissolving/ageing tank; and

3) Stock tank.

(iii) If dissolving tank and stock tank are set at the same level or gravity transfer is not possible, the system shall incorporate a suitable transfer pump;

(iv) Where full standby provision is required, the packaged unit comprising charging, storage, feeding and wetting unit, dissolving/ageing tank and stock tank shall be duplicated; and

(v) Where pumps are required to transfer polyelectrolyte from dissolving/ageing tank to stock tank, these shall be duplicated and shall be capable of operating with any combination of tanks.

(d) Polyelectrolyte Concentration

(i) The polyelectrolyte plant shall be designed to prepare stock solution of polyelectrolytes at concentrations up to those recommended by the product manufacturer. The solution shall be diluted 5 to 10 fold for dosing purposes.

(e) Polyelectrolyte Handling and Charging Systems

(i) Refer to the section on Solid Chemical Unloading and Storage.

(f) Feeding and Wetting Devices

(i) Heated pneumatic conveying shall be used to transfer the powder from feeder to the ejector; and

(ii) The feeding unit shall incorporate a screw feeder for accurately metering the polyelectrolyte to the pneumatic conveyance system. The screw feeder shall operate under control of an adjustable timer to provide a means of varying the batch concentration. The reproducibility of the feeder shall be better than + or - 3% by weight.

(g) Mixing / Ageing Tanks

(i) Mixing/ageing tanks shall be designed for receipt of polyelectrolyte from a suitable particle wetting device. The polyelectrolyte shall be dissolved/diluted to form a solution of uniform concentration and consistency, free of lumps and 'fish-eyes' and allowed to age before being discharged.

(ii) The tanks shall be sized such that the required stock volume can be produced in a single batch at the specified concentration.

(iii) The tanks shall be fabricated from stainless steel Type 316S13 to BS EN 10029, BS EN 10259, BS EN 10258, BS EN 10048, BS EN 10095, BS EN 10051 or other suitable corrosion and abrasive resistant suitable material approved by the S.O.. The tanks shall generally be cylindrical and suitable for mounting on a flat concrete plinth. Alternatively, the tanks may be mounted on a platform above the stock tanks or on top of the stock tank. Mild steel tanks shall not be used.

(iv) The tanks shall be located within a bunded area inside a building. The bund shall be dedicated to polyelectrolyte handling and shall be designed to accommodate 110% of the contents of the largest tank contained therein.

(v) Each tank shall be equipped with an electric medium speed propeller type or similar mixer operating at 400-500 rpm. The shaft and propeller shall be fabricated from stainless steel type 316S13 to BS EN 10095, BS EN 10250-4, BS EN 10085, PD 970, BS EN 10087, BS EN 10083-1, BS EN 10084 or BS EN 10029, BS EN 10259, BS EN 10258, BS EN 10048, BS EN 10095, BS EN 10051. The mixer shall operate under control of an adjustable timer.

(vi) Where applicable, suitable level switches or probes shall be provided for operational purposes including low level, initial fill level, high level and overflow level.

(vii) Where specified in the Particular Requirements, mixing/ageing tanks may also serve as stock tanks subject to the further requirements of the ensuing clauses.

(h) Stock Tanks

(i) Stock tanks shall be designed for receipt of polyelectrolyte solution from a mixing/ageing tank and/or where specified, liquid polyelectrolyte from drums. The capacity of the stock tank shall be greater than or equal to the capacity of the dissolving tank and shall be sized to provide storage at the maximum demand over the time taken to prepare a batch.

(ii) The stock tank could be an integral part of the mixing/ageing tank.

(iii) The tanks shall be fabricated from stainless steel type 316S13 to BS EN 10029, BS EN 10259, BS EN 10258, BS EN 10048, BS EN 10095, BS EN 10051or other suitable corrosion and abrasive resistant material approved by the S.O.. The tanks shall be generally cylindrical and suitable for mounting on a flat concrete plinth. Mild steel tanks shall not be used.

(iv) The tanks shall be located within a bunded area inside a building. The bund shall be dedicated to polyelectrolyte handling and shall be designed to accommodate 110% of the contents of the largest tank contained therein.

(v) Where applicable, suitable level switches or probes and level measuring equipment shall be provided for operational purposes including extra low level, low level and overflow level and for monitoring purposes.

(i) Dilution System

(i) Polyelectrolyte solution shall be diluted prior to injection to improve mixing. The dilution ratio shall generally be as per specified.

(j) Pipework

(i) Stainless steel pipework shall be type 316S13 to BS EN 10216 and BS EN 10217 and shall be flanged. Fittings shall comply with BS EN 10253 Part 4. All welding shall be carried out to BS 4677 and procedures shall be subject to approval in accordance with BS EN 288 3. Welders shall be tested by an independent testing authority and shall satisfy the requirements of BS EN 287 1;

(ii) Welder test certificates shall be provided by the Contractor for examination by the S.O.;

(iii) All welds shall be dressed;

(iv) uPVC pipework shall be as specified above;

(v) Reinforced plastics shall be as specified above;

(vi) Alternative plastic materials such as polypropylene and polyvinylidene difluoride (PVDF) may be used in certain circumstances subject to approval by the S.O.; and

(vii) Joint rings for steel pipework shall be proprietary and approved by the S.O..

(k) Isolating Valves

(i) General

1) All isolating valves shall be positioned with due regard to routine access for operational and maintenance purposes. It shall not be necessary to enter any bunded facility to operate isolating valves; and

2) Isolating valves shall be compatible generally with the pipework systems in which they are installed.

(ii) uPVC Valves

1) Isolating valves shall be of the diaphragm type with uPVC bodies, glass filled polypropylene top works and PTFE diaphragms. Assembly nuts and bolts shall be stainless steel SS316L;

2) End connection fittings may be plain sockets for solvent welding or union coupling type or flanged as appropriate. Flanges shall be to BS EN 1092-1 with minimum PN10 or PN 16 where appropriate; and

3) These valves may not be used for isolating tank outlets and drains unless otherwise approved by the S.O..

(iii) Rubber Lined Valves

1) Isolating valves shall be of the diaphragm type to BS EN 13397 with weir bodies in grey cast iron to BS EN 1561 grade 220 (or alternative grade allowed by the design standard) having integral flanged ends, a natural rubber ebonite lining, PTFE diaphragms and top works in standard proprietary materials. Assembly nuts and bolts shall be in stainless steel. The valves shall be flanged to BS EN 1092- with minimum PN10 or PN 16 where appropriate; and

2) Isolating valves for tank outlets shall be of the sealed bonnet type.

G4.47.29 Carbon Dioxide Plant

(a) General

(i) In additional to the general requirement, this Section contains requirements which, where relevant to the Contract, shall apply to the supply, installation and testing of Carbon Dioxide plant.

(ii) The carbon dioxide plant shall consist of storage tanks, refrigeration units, vaporisers, metering and dosing units, leak detectors, all interconnecting pipework, valves and fittings.

(iii) The complete installation shall conform as a minimum to the following reference standards (latest edition) as applicable:

1) Bulk Storage of Liquid Carbon Dioxide: Hazards and Procedures, Guidance Note CS 9, Health and Safety Executive, UK;

2) Standards for Low Pressure Carbon Dioxide Systems at Consumer Sites, CGA G-6.1, Compressed Gas Association, US; and

3) Potential liquid carbon dioxide suppliers' technical guidelines.

(iv) In selecting a location for liquid carbon dioxide tank and pipework the Contractor shall ensure that adequate safety distances are allowed between any point on the liquid carbon dioxide system where in normal operation a carbon dioxide release can occur and areas occupied by personnel or ducts, pipe trenches and ventilation plant intakes which could transmit leaks to such areas.

(v) Tanks shall not be located below ground or near cellars or underground rooms, where carbon dioxide can collect. All carbon dioxide plant areas shall be well ventilated. All enclosed and confined areas provided with man access where carbon dioxide can accumulate shall be provided with mechanical ventilation and shall be monitored with continuous carbon dioxide analysers. Analysers shall indicate carbon dioxide concentrations at the entrances and shall activate high concentration visual and audible alarms. Liquid carbon dioxide plant where possible shall be restricted to outdoor location.

(vi) A reputable liquid carbon dioxide supplier shall be used for the design, construction, procurement, testing, installation and commissioning of the liquid carbon dioxide system consisting of storage tanks, vaporisers, pipework, valves and fittings.

(vii) Proof shall be submitted with the tender that the proposed supplier has been engaged in the specified work on plants of similar capacity for at least 10 years.

(b) Liquid Carbon Dioxide Storage Tanks

(i) Storage tanks shall be either vertical or horizontal as specified;

(ii) The tanks shall be of electrically welded carbon steel construction with dished ends and all required flanged nozzles made in one plate;

(iii) The tanks shall be designed for a pressure of 24 bar gauge and a temperature range of -80 ^{0}C to +50 ^{0}C and shall be suitable for operating conditions of 20.3 to 21.0 bar gauge at -17 ^{0}C to -16 ^{0}C. The tanks shall be pressure tested initially to 33.1 bar gauge;

(iv) The tanks shall be designed, constructed, tested and inspected in accordance with BS PD 5500: Category 2. Materials shall be to BS 1501-224-490A-LT50, impact tested to BS EN 10028-LT50. The tanks shall be stress relieved by heat treatment after manufacture, all welded seams first having been radiographed;

(v) Each tank shall be provided with an 800 mm diameter davit and blank flange for manway and branches for making necessary connections for bottom (liquid) and top (gas space) filling pipes and outlet, coolant inlet and outlet, pressure relief and instrument carbon dioxide supply;

(vi) Contents indication shall be continuous by differential pressure measurement and shall be displayed in tonne on a dial type indicator or digital indications of equal readability showing both % full and tonnage mounted in front of the tanks at the filling point and on the SCADA system;

(vii) The tanks shall be of double wall construction consisting of an inner tank supported within an outer tank;

(viii) The annular space between the inner and outer tanks shall be filled with multiple layered laminate Perlite or powdered Perlite insulation and sealed under vacuum. It shall be provided with a vacuum filter and valve, a thermocouple to monitor temperature and relief valve;

(ix) The complete storage tank shall be designed to limit evaporation losses due to heat gain and pressure regulation to not more than 0.25 percent by weight per day of the storage capacity;

(x) The pressure relief shall be by four high capacity safety relief valves, each set to lift at 24.1 bar g. The relief valve shall be mounted on two, 3 port, 2-way change over unit to ensure that the vessel is always protected by at least two relief valves. Safety valves with hand lift levers for testing shall not be used;

(xi) The tanks shall be provided with filling pipes, to top and bottom of the tank, fill connection, pressure relief valves between isolating valves in pipework where liquid carbon dioxide could get trapped, groundings for dissipation of lightning strikes and static electrical discharges and all other safety and operating devices;

(xii) The tanks shall be supported on a steel base frame which shall form part of the tanks. The tank supports shall be designed for wind loading from wind velocities experienced on site; and

(xiii) Lugs shall be provided for lifting the tanks when empty.

(c) Refrigeration Unit

(i) The refrigeration system shall be of the air cooled type and shall be attached to the storage tank and located in an aluminium control housing. The system shall automatically maintain carbon dioxide in the storage tank at $-17\ ^{0}$ C to $-16\ ^{0}$C and 20.3 to 21.0 bar g. The refrigeration unit shall be equipped with condensing unit and provided with a fused disconnect switch, a motor starter and a control voltage transformer. The condensing unit shall include a sight glass, refrigerant pipe, solenoid valve, expansion valve, refrigeration coil mounted in the carbon dioxide storage tank and a heat exchanger. Automatic control shall be provided to start/ stop the compressor, thereby controlling the temperature of carbon dioxide and maintaining the proper operating pressure;

(ii) The unit shall be designed to operate on an ozone friendly refrigerant;

(iii) The unit shall be mounted on a steel framework and shall be suitable for installation both indoors and outdoors;

(iv) The refrigeration unit shall be controlled by pressure switches to maintain the designed working pressure in the storage vessel and shall be supplied complete with refrigerant pipework; and

(v) The malfunction of the refrigeration unit shall be monitored by using the pressure alarm on the storage vessel.

(d) Vaporisers

(i) Ambient Heat Type

1) Ambient heat vaporisers shall be capable of receiving liquid carbon dioxide at – 17 ^{0}C to – 16 ^{0}C and 20.3 to 21.0 bar gauge from the storage vessel and continuously converting it to gaseous form. They shall be operated by natural air flow. Alternatively, air shall be drawn through the heat exchanger element by fans at a uniform flow velocity;

2) Vaporisers shall consist of heat exchangers with elements made up of finned copper or aluminium tubing to maximise heat transfer. Each vaporiser shall be designed for at least continuous 8 hours operation at the maximum demand without frosting. Power operated valve controls shall be provided for automatic changeover from duty to standby vaporiser on a timed basis to prevent vaporiser frosting. Any condensate from the tubing shall be collected in a tray provided with a drain;

3) A low temperature cut off switch shall be installed downstream from each vaporiser to automatically isolate carbon dioxide supply to safeguard the downstream equipment from the effects of liquid carbon dioxide carryover;

4) The vaporisers shall be suitable for outdoor installation; and

5) The vaporisers shall be designed and tested to the maximum liquid carbon dioxide storage vessel operating pressure.

(ii) Electrically Heated Type

1) Electrically heated vaporisers shall be thermostatically controlled, and suitable for converting liquid carbon dioxide at - 17 ^{0}C to - 16 ^{0}C and 20.3 to 21.0 bar gauge continuously to gaseous carbon dioxide at an exit temperature between 15 ^{0}C and 30 ^{0}C;

2) The heating elements shall be thermally insulated. The piping used for carbon dioxide in the vaporisers shall be solid drawn carbon steel tubing to BS EN 10216 or BS EN 10217 as applicable;

3) Vaporisers shall be fully protected against excess temperature, pressure and electrical loads and shall be provided with a temperature sensor arranged to automatically close a power operated valve on the inlet to stop any flow of liquid carbon dioxide reaching the downstream equipment. Each unit shall be provided with a panel mounted ammeter, power supply ON/OFF indicator and carbon dioxide discharge temperature gauge;

4) Vaporisers shall be self regulating; and

5) The vaporisers shall be designed and tested to the maximum liquid carbon dioxide storage vessel operating pressure.

(e) Carbon Dioxide Liquid and Gaseous Pipework

(i) Liquid Pipework

1) The liquid carbon dioxide pipework system shall be designed to ANSI B 31.3;

2) The pipe material shall be austenitic stainless steel to BS EN 10216 and BS EN 10217 type 304S18, 316S18 or 321S18 with flanges welded on. Flange material shall be ASA Class 300 from BS EN 10222 type 304S31, 316S31 or 321S31;

3) Bolt material shall be to BS 4882 grade L7 and nut material shall be to BS 4882 grade L4;

4) Manual metal arc welding shall be used for rolled pipe butt and slip on flange welds. TIG process shall be used for small bore pipework; filter wire material shall be selected from BS EN ISO 14343;

5) Welding procedures and welders working to approved welding procedures shall comply with the requirements in BS EN ISO 14732 and BS 4871 Part 3 and details shall be submitted for approval before fabrication;

6) Following welding work on the pipework the entire section of the pipework shall be tested by magnetic particle flow detection to BS 6072 or by radiography as applicable, bolt material to BS 4882 grade L9 and nut material to BS 4882 grade L4;

7) The whole of the pipework shall be supported with purpose made brackets and hangers. All pipework carrying liquid carbon dioxide shall be:

- Anchored to accommodate shrinkage of the piping due to passage of the cold liquid carbon dioxide through them;
- Lagged, insulated and finished to the same specification as for the storage vessels;
- Earthed to eliminate the build up of static electricity potential; and
- Provided with safety relief devices to prevent excessive pressure build up from entrapped liquid between isolating valves.

8) All screwed pipework conveying carbon dioxide gas under pressure shall be constructed from solid drawn steel tube to BS EN 10255 with heavy fittings in malleable iron to BS 143 screwed to BS EN 10226-1. Threaded joints shall be sealed using a jointing compound compatible with carbon dioxide; and

9) All flanged pipework conveying carbon dioxide gas under pressure shall be carbon steel hot finished seamless, cold drawn seamless or electric resistance, welded to BS EN 10216 and BS EN 10217 grade 360 or 430. Flanges shall be forged carbon steel BS EN 10222-224-490 raised face drilled to BS EN 1759 Class 300, socket welded, stress relieved and radiographed. Joint rings shall be chemical resistant elastomers such as Hypalon (chloro-sulphonated polyethylene) or Viton (co-polymer of vinylidene fluoride and hexafluoro propylene) or alternative approved by the S.O. and shall fit within the bolt circle. Wire reinforced material shall not be used. Prior to installation, the joint rings shall be impregnated or smeared with a compound compatible with carbon dioxide. Graphite shall not be used.

(f) Gaseous Pipework

(i) Gaseous pipework shall be provided with safety relief devices to prevent excessive pressure build up due to failure of pressure reducing valves;

(ii) When relief valves are located in enclosed rooms, the outlet from the valve shall be piped to the outside. Such piping shall not be equipped with valves, traps or any other means of stopping or restricting the gas flow. Venting pipework shall not be manifolded. The horizontal run of vent pipes shall be on a continuous down gradient with the outer end turned down and fitted with an insect screen;

(iii) All pipework for carbon dioxide gas under vacuum or venting directly to atmosphere shall be rigid 'hi-impact' uPVC to BS 3506 Class E or T with solvent welded joints and fittings complying with the relevant parts of BS EN 1452;

(iv) Pipework shall be adequately supported throughout its length. Sleeves shall be used where pipework passes through walls. All apertures shall be properly sealed with carbon dioxide resistant compound;

(v) Pipework within buildings shall not be run in floor ducts unless otherwise approved by the S.O.; and

(vi) All pipes and pipework assemblies for carbon dioxide duty shall be hydraulically tested to 1.5 times the maximum operating pressure.

(g) Valves for Carbon Dioxide

(i) Valves for liquid carbon dioxide duties shall be austenitic steel grade 316C16 to BS EN 10213, flanged ball valves complying fully with BS EN 1561;

(ii) The ball shall be of one piece construction not coated in any way, screwed-insert to be retained via one end only of one piece body. The stem shall be inserted from inside the body;

(iii) Seat material shall be virgin PTFE;

(iv) Flanges shall be fully machined to BS EN 1759, class 150 (or ANSI 150 B 16.5) standard finish on raised face surface;

(v) Bolt material shall be to BS 4882 grade L7 and nut material shall be to BS 4882 grade L4;

(vi) Valves shall be pressure tested to BS EN 12266;

(vii) Valves in pipework carrying carbon dioxide gas under pressure shall be specifically designed for carbon dioxide service and shall comprise a forged carbon steel body with integral end connections, tapered plug in Monel with pure PTFE sleeve and top seal arrangement with PTFE diaphragm, PTFE delta seal ring, back up metal diaphragm, floating thrust collar, electrostatic eliminator, four bolt cover and three point self-aligning plug adjustment. The valve shall be operated by a forged steel wrench with padlocking facility in the closed position or otherwise by power actuator as per specified. The valve design shall ensure that excess pressure in the plug and body cavity of the closed valve is relieved spontaneously toward the right direction. The valve shall be supplied degreased and dried for carbon dioxide service and delivered packed and labelled accordingly;

(viii) Relief valves shall be of the spring loaded type. The valve body shall be manufactured in brass with a stainless steel seat and spring and disc manufactured in grade 316S18 to BS EN 10216 and BS EN 10217;

(ix) Valves in pipework carrying carbon dioxide gas under vacuum shall generally be of the ball type with uPVC bodies and FPM seals. Manual isolating valves for carbon dioxide solution duties shall generally be of the diaphragm type with uPVC bodies and diaphragm resistant to carbon dioxide solution;

(x) Assembly nuts and bolts shall be stainless steel. End fittings may be plain sockets for solvent welding or flanged; and

(xi) Motorised valves shall be uPVC ball vales.

(h) Gas shut off Valves

(i) Gas shut off valves shall ensure quick automatic shut off of the gas on receipt of a signal from a controller. The opening of the valve following automatic shut off shall be manual by push button or similar; and

(ii) The shut off valve actuator shall be pneumatically operated using air or carbon dioxide gas supply from the plant or electrically operated and shall be arranged to fail safe in the closed position in the event of an electrical or gas supply failure.

(i) Pressure reducing Valves

(i) Pressure reducing valves shall be heavy duty spring loaded diaphragm valves designed to regulate varying upstream gas pressure automatically to a desired downstream pressure. The flow operating range of the valve shall be at least 100:1;

(ii) Valves shall be provided with positive manual gas shut off facilities; and

(iii) A filter unit shall be provided to protect the pressure reducing valve from impurities in the gas. The filter unit could either form an integral part of the valve or be separate.

(j) Vacuum Regulating Valves

(i) Vacuum regulating valves shall be wall mounted in the evaporator room located as close as is practical downstream of the evaporators to minimise the length of gas pipework under pressure;

(ii) Vacuum regulating valves shall be spring loaded diaphragm operated valves, designed to reduce varying gas pressure upstream to a desired regulated vacuum downstream and to maintain this vacuum within close limits. The diaphragm shall be made of PTFE;

(iii) The vacuum regulating valve shall be fail safe and shall close fully upon loss of vacuum;

(iv) The valve shall be provided with an isolating valve upstream and a pressure relief valve downstream. The pressure relief valve shall discharge to outside the building; and

(v) The regulated vacuum shall be indicated on a vacuum pressure gauge which may be integral with the vacuum regulating valve or mounted separately.

(k) Gas Flow Regulating Valves

(i) Gas flow regulating valves shall be globe valves constructed of stainless steel 316C16 to BS EN 10213;

(ii) Valve spindle shall be stainless steel type 316S16 to PD 970;

(iii) Gland packing and seating ring shall be PTFE;

(iv) Valves shall be sized in accordance with BS EN 60534 – 2; and

(v) Flanged valves shall have integral flanges drilled to BS EN 1092 with pressure designation to match the pipework system. The flow control range of the valves shall be 10:1 or greater.

(l) Carbon Dioxide Feeders

(i) Carbon dioxide feeders are referred to herein as gas control units. They are devices for measuring and controlling the flow rate carbon dioxide gas under vacuum conditions created by a water driven ejector;

(ii) Gas control units shall be wall mounted unless specified otherwise and shall incorporate the following items all assembled and tested at place of manufacture;

1) Variable area flow meter with linear scale calibrated in g/s or mg/s of chlorine as applicable; and

2) Gas flow control valve to provide precise setting of the gas flow rate over a 20:1 flow range with an accuracy of ±4% of the indicated gas flow rate.

(iii) Where automatic operation is required, the gas flow control valve shall be fitted with a motorised actuator and positioner utilising a 4-20 mA DC signal from a remote controller. The actuator and positioner shall be suitable for a 230 V AC power supply unless specified otherwise. A manual mechanical override shall be provided;

(iv) The gas control unit shall have an auto/manual control selector switch. In manual mode, the gas flow rate shall be set by hand at the unit itself. In automatic mode, the unit shall be controlled from a remote controller. The switch shall have auxiliary volt free contacts connected to the SCADA system to allow the selected position remotely;

(v) On power failure, the gas flow control valve shall stay put; and

(vi) Other components will be as follows:

1) Vacuum differential regulating valve to maintain a constant vacuum difference between the inlet and the outlet of the flow control valve over the operating range of the gas control unit;

2) Vacuum relief valve, if applicable, which shall be connected by pipework to outside the building and arranged in a similar manner to venting pipework;

3) Vacuum gauge to indicate the unregulated vacuum induced by the ejector and calibrated in mbar; and

4) Vacuum switches, which may be integral with the gauge. One switch shall initiate a high vacuum alarm and the other shall initiate a low vacuum alarm. Both switches shall have volt free changeover contacts.

(m) Ejector

(i) Ejectors shall be designed to induce the vacuum necessary to operate the corresponding gas control units over their working range under all downstream pressure conditions at the point of application when supplied with motive water at or above the minimum pressure;

(ii) The design flow range of motive water shall consider the dissolution requirements of the chemical, control loop time considerations and effective hydraulic design. Noise shall be kept to an absolute minimum consistent with efficient operation but shall in no event exceed the sound pressure level stated in the General Specification;

(iii) The ejector shall incorporate an integral diaphragm operated non return valve to prevent ingress of water to the gas system. A separate non return valve shall also be installed in the gas line to the ejector as a security measure;

(iv) The nozzle and throat diameters shall be engraved on the plate fixed to the ejector in a visible position. Where small clearances are involved on the water side of the ejector, in line strainers shall be installed in the motive water supply pipework to protect the ejectors from blockage; and

(v) Ejector shall be mounted separately from the gas control units either within the same room or in a remote location as per specified.

(n) Solution Pipework

(i) All pipework within buildings, unless otherwise specified, shall be rigid 'hi-impact' uPVC to BS 3506 Class E with solvent welded joints and fittings complying with the relevant parts of BS EN 1452;

(ii) Where flanged joints are required, full face uPVC flanges and stainless steel SS316L backing rings shall be used. Joint rings shall be Hypalon or Viton. Flanges shall be drilled to BS EN 1092 PN10 minimum; and

(iii) Pipework shall be supported in accordance with manufacturer's recommendations and adequate provision shall be made for thermal expansion and contraction.

(o) Solution Isolating Valves

(i) Manual isolating valves for carbon dioxide solutions shall be of the diaphragm type with uPVC bodies, glass filled polypropylene top works and PTFE diaphragms; and

(ii) Assembly nuts and bolts shall be stainless steel SS316L.

(p) Motive Water Pumps

(i) Motive water pumps shall comply with the General Specification Pumps.

(q) Side Stream Mixer

(i) Side stream mixers shall be static mixers as specified in General Specification Mixing and Flocculation Systems.

(r) Carbon Dioxide Solution distribution and Injection Fittings

(i) Where carbon dioxide is to be dosed into flow in an open channel, weir chamber, and contact tank or downstream of a hydraulic jump, it shall be applied using an ammonia solution distributor; and

(ii) The distributor shall be either a drilled tube or a ceramic tubular diffuser, designed to ensure uniform distribution of solution at the point of application.

(s) Where carbon dioxide solution is to be dosed into flow in pipelines, it shall be applied using an injection fitting/device designed for the specified duty flowrate.

(t) Injection Pipework configuration shall be as follows:

(i) For pipelines up to 600 mm diameter, the injection tube shall extend one-third of the pipeline diameter into the fluid stream. In this case a withdrawable type injection fitting entering the pipe through a corporation cock shall be used;

(ii) For pipelines up to 1000 mm in diameter, the injection tube shall be perforated and shall extend right across the pipe bore and be supported with ends located in diametrically opposite flanged branches. The tube shall be drilled at predetermined centres to ensure uniform distribution across the flow profile;

(iii) For pipelines up to 2500 mm in diameter, two perforated injection tubes shall be used, installed mutually at right angles and with their axes at 45^0 to the horizontal in a plane normal to the direction of flow within the pipeline;

(iv) Alternatively, four injection nozzles, mutually located at 90^0 and extending 15% of the pipe diameter shall be installed. The nozzles shall be installed in diametrically opposite flanged branches and in the same plane normal to the direction of flow in the pipeline;

(v) Where a static mixer (or motionless mixer) is to be used for mixing carbon dioxide solution, the injection fitting/device shall be designed, and located in accordance with static mixer supplier's requirements; and

(vi) The materials of construction of the distributors, injectors and sealing shall be compatible with up to 2500 mg/l carbon dioxide solutions at operational fluid temperatures up to 30 ^{0}C. Porous ceramic material where used shall be inert and non-toxic. UPVC tube where used shall conform to BS 3506, Class E.

(u) Self-contained Air Breathing Apparatus

(i) Self-contained air breathing apparatus shall comply with the requirements in the specification for the 'Safety Equipment'.

G4.47.30 Inspection and Testing

(a) Manufacturer's Inspection

(i) An experienced, competent, and authorised representative of the equipment manufacturer shall visit the fabricator's work site and inspect, check, adjust if necessary, and approve the equipment skid fabrication;

(ii) The representative shall be present when equipment is tested in accordance with the requirements herein;

(iii) The representative shall reserve the right to reject equipment or skid fabrication if it is determined to be inappropriate or inadequate;

(iv) The representative shall revisit the jobsite as often as necessary until all defects are corrected and the equipment installation and operation are satisfactory;

(v) The system supplier's representative shall furnish a written report certifying that the skids and pumps have been properly fabricated; that piping has been properly cleaned; that equipment and piping are in proper alignment and free from undue stress imposed by connecting piping or anchor bolts; and that the skids and pumps have been tested according to the requirements herein; and

(vi) All costs for these services shall be included in the contract price for the number of days and round trips to the site as required.

(b) Board's Witness Inspection and Testing

(i) Following the Manufacturer's Inspection, each pump skid shall be inspected and witness tested by the Board's representative prior to shipment. The S.O. reserves the right to reject any equipment and/or skids if it is determined that it has failed the shop testing or if incorrect pipe materials, incorrect valve types, incorrect solvent cements, shoddy construction or equipment malfunction are recognized.

(c) Shop Testing

(i) As stipulated herein, each pump skid and pump shall be tested with water (over the full capacity range of pumping requirements) and for pressure capabilities. Each pump shall be checked with calibration columns for low and high end flow capabilities. Pumps shall be tested with the motors to be provided. Board or Board's representative shall witness shop tests, inspect and check the testing equipment used, and observe the calibration of pressure gauges and flow meters;

(ii) For witnessed shop tests, all readings are to be read manually from the certified and/or calibrated instruments. The use of computer data acquisition systems shall not be acceptable; and

(iii) If a pump or skid fails to operate properly or fails to meet the specified conditions or requirements during witnessed shop testing, the skid fabricator shall modify the pumping unit and perform re-testing.

(d) Calibration Graph

(i) The Contractor shall prepare a calibration graph from the shop tests for each chemical feed metering pump which does not have a rate set device reading in litres per hour of liquid; and

(ii) The graph shall show the rate setter graduation conversion to litres per hour throughout the full range of the feed unit. Each graph shall be furnished on hard paper and sealed in clear plastic.

G4.47.31 Testing of Chemical Plant

(a) Offsite Inspection and Testing

(i) Test shall be carried out on all items of the plant, vessels, pipework and valves to demonstrate compliance with the criteria specified and with relevant standards and test specified; and

(ii) Unless otherwise specified, the tests shall be in accordance with the Reference Standards and the manufacturers' own procedures.

(b) Individual Tests

The following inspection and test shall be carried out as appropriate:

(i) Inspection to check the assembly of the plant and conformity with the Specification and approved Drawings;

(ii) Rotational checking of all electric motors;

(iii) Functional testing of all auxiliary items including mixers and valve actuators;

(iv) All tanks shall be filled to overflow level and inspected for absence of leaks;

(v) Hydrostatic testing of all pipework systems at 1.5 times maximum working pressure for a period of at least one hour; during this time the pressure shall not change;

(vi) Relief valve operating pressure shall be demonstrated;

(vii) Pressure reducing and shut off valve operation shall be demonstrated; and

(viii) Functional testing of each pump to prove correct operation, absence of fluid leaks, correct gearing and gear box temperatures and absence of undue vibration and noise.

(c) Preliminary Tests

Testing shall incorporate the following requirements:

(i) Functional testing (without the chemical) of the chemical delivery, loading and batching system to demonstrate correct operation with respect to sequence, time and levels;

(ii) Functional testing of the batching system with the respective chemical to demonstrate correct operation and calibration;

(iii) Where required, functional tests shall be performed on the solids handling systems with demonstration of correct operation of safety devices including silo pressure relief valves, all prior to charging with chemical;

(iv) Where required, functional test shall be performed on solids handling systems with chemical to demonstrate correct operation and calibration when handling the chemical; and

(v) Dosing pump calibration shall be demonstrated and specified accuracies shall be confirmed.

(d) The following additional requirements for the testing of hydrated lime plant shall apply:

(i) Individual Testing

1) For hydrated lime plant, accuracy of primary lime tank weighers shall be demonstrated with water.

(e) The following additional requirements for the testing of carbon dioxide plant shall apply:

(i) Offsite Inspection and Testing

1) Fans shall be tested in accordance with the relevant sections in BS 848: Parts 1 and 2. Pressure vessels shall be tested in accordance with BS 5500 or ASME Pressure Vessel Code, Section VIII, Division 1;

2) Welders of Carbon dioxide plant and pipework shall be tested and approved in accordance with the requirements of the appropriate parts of BS 4871;

3) Non-destructive tests including radiography and / or other methods as specified shall be carried out on all welds in plant and pipework;

4) All liquid and gas isolating valves shall be tested in accordance with BS 6755 Part 1;

5) All liquid handling plant shall be subjected to hydrostatic pressure hold test at 1.5 times the design pressure for at least one hour;

6) All gas handling plant shall be subjected to the following tests;

7) Vacuum hold test at –1.0 bar g for 1 hour where items normally operate under vacuum;

8) Pressure hold test at 1.5 times the design pressure for at least one hour using air or nitrogen. Where items normally operate under vacuum, a pressure hold test at a pressure of at least 1.5 bar g or the highest pressure that can occur under fault conditions whichever is greater shall be carried out;

9) Functional tests shall be carried out on all vaporisers, pressure reducing / shut off valves, pressure relief valves, gas flow controllers, ejector static mixers and motive water pumps which shall be temporarily rigged as systems to demonstrate correct operation and calibration. Water supply pressure and solution back pressure for these tests shall be set to the limiting values anticipated at Site;

10) Functional tests shall be carried out on any packaged plant (fully assembled) to demonstrate correct operating sequence; and

11) Type test certificates shall be provided for the particulate removal filters on the plant.

(ii) Individual Tests

1) Liquid pipework system up to the vaporisers shall be tested using dry bottled air or nitrogen for one hour at 1.5 times maximum working pressure at 40 ^oC. During this time pressure shall not change. Where necessary, system pressure gauges may be removed prior to testing. On re-instatement of the gauges the system shall be re-tested at gauge maximum reading.

2) The gas pipework system from the vaporiser outlet up to the ejector inlet connections shall be tested for at least one hour at 1.5 x maximum working pressure at 40 ^oC using dry air or nitrogen. During this time the pressure shall not change. Where necessary, system pressure gauges may be removed prior to testing. On re-instatement of the gauges, the system shall be re-tested at the gauge maximum reading.

3) The carbon dioxide solution pipework including ejectors, static mixers and all components up to the point of application shall be tested hydrostatically for at least one hour at 1.5 x maximum working pressure or 5 bar g whichever is greater. Manufacturers' recommended procedures shall be followed prior to execution of this test with particular reference to:

- Elimination of air on filling with cold water;
- Temperature equilibration; and
- Stepwise application of test pressure starting at not more than 3 bar (g) for 10 minutes.

(iii) Preliminary Tests

1) Carbon dioxide storage tank operating in conjunction with the refrigeration unit shall be tested to demonstrate that there is no loss of carbon dioxide from the tank. The Contractor shall submit a test procedure to verify 'no loss of carbon dioxide' and shall monitor time, carbon dioxide storage tank liquid level and pressure and running time of the refrigeration unit for a period of 10 days with the tank idle (no draw off) with readings taken every 12 hours.

2) Vaporiser shall be tested for continuous vaporisation cycle at the maximum vaporisation rate required and to demonstrate that the duration of the defrosting period achieve the specified 8 hour period. Test data shall indicate as a minimum time, ambient temperature, and carbon dioxide gas flow rate, gas pressure at regulating valve outlet, gas temperature between vaporiser and regulating valves, vaporising and defrosting cycle times and liquid carbon dioxide tank pressure, level and temperature. A minimum of three complete vaporising / defrosting cycles shall be tested with reading recorded every hour.

3) Side stream static mixer shall be tested for mixing of carbon dioxide in water.

4) Functional tests with carbon dioxide shall be carried out to demonstrate correct operation and calibration of the plant.

END OF SECTION